

SingleRAN

Equipment Security Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised the technical description of key file integrity monitoring. For details, see [Technical Description](#).

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 05 (2025-12-14).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added the support for functions including secure boot by the UMPThb series main control boards. For details, see: 4.1.1.2.2 Restrictions on Application Scenarios 4.3.3 Hardware 6.3 Base Station Security Environment	None	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the support for obtaining CRLs over HTTP. For details, see 4.1.1.2.3 Automatically Configured ACL Rule Group.	Modified parameter: Added the enumerated value HTTP to the CRLTSK.CRLGETMET HOD (LTE eNodeB, 5G gNodeB) parameter.	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR. This feature works the same way with each of these RATs.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RA T	Feature ID	Feature Name	Chapter/Section
GS M	MRFD-210 102	Operate System Security Management	6.2.2 OS Security of the BSC6900/ BSC6910 (GSM/ UMTS)
UM TS	MRFD-210 102	Operate System Security Management	
GS M	MRFD-210 305	Security Management	6.3 Base Station Security Environment 6.4 Base Station Self-Check Upon Startup 6.5 Base Station Process Auditing
UM TS	MRFD-210 305	Security Management	
LTE FD D	LBFD-0040 10	Security Management	
LTE TD D	TDLBFD-00 4010	Security Management	
LTE FD D	LOFD-0030 14	Integrated Firewall	4 Integrated Firewall
LTE FD D	LOFD-0030 1401	Access Control List (ACL)	
LTE FD D	LOFD-0030 1402	Automatic ACL Rule Configuration	
NB- IoT	MLOFD-00 3014	Integrated Firewall	
NB- IoT	MLOFD-00 301401	Access Control List (ACL)	
NB- IoT	MLOFD-00 301402	Automatic ACL Rule Configuration	
LTE TD D	TDLOFD-0 03014	Integrated Firewall	
LTE TD D	TDLOFD-0 0301401	Access Control List (ACL)	

RA T	Feature ID	Feature Name	Chapter/Section
LTE TD D	TDLOFD-0 0301402	Automatic ACL Rule Configuration	
NR	FBFD-0100 23	Security Mechanism (gNodeB Supporting Integrated Firewall)	4 Integrated Firewall

3 Overview

Table 3-1 lists the equipment security measures supported by Huawei SingleRAN.

Table 3-1 Supported security measures

Security Measure		Base Station Controller (GSM/UMTS)	Base Station
Physical security		√	√
Operating system (OS) security	OS hardening	√	√
	OS patches	√	√
Security environment		x	√
Self-check upon startup		x	√
Process auditing		x	√
Key file integrity monitoring		√	√
Integrated firewall		√	√
Physical port security		√	√
NOTE √ indicates that the NE supports this security measure. x indicates that the NE does not support this security measure. - indicates that the NE does not involve this security measure.			

 NOTE

In this document, MBSC is referred to as the BSC6900/BSC6910/BSC6960, and eGBTS, NodeB, eNodeB, gNodeB, and MBTS are collectively referred to as the base station.

Regarding the integrated firewall of the GBTS, see [4 Integrated Firewall](#) in this document. For other equipment security measures, see *GBTS Equipment and OM Security* in GBSS feature documentation.

4 Integrated Firewall

4.1 Principles

4.1.1 Integrated Firewall of Base Stations

Base stations deployed on networks may encounter various network attacks. Therefore, lightweight firewalls are integrated into base stations to support ACL-based packet filtering, flood attack prevention, invalid packet attack prevention, and other functions.

4.1.1.1 ACL-based Packet Filtering

On the base station side, ACL rules are used to filter invalid Layer 2 (data link layer), Layer 3 (network layer), and Layer 4 (transport layer) packets. ACL-based packet filtering involves ACLs and filtering actions. ACL-based packet filtering is supported in both IPv4 and IPv6 networking scenarios. In IPv4 networking, both old and new transmission configuration models are supported. The model to be used is specified by the **GTRANSPARA.TRANSFCGMODE (5G gNodeB, LTE eNodeB)** parameter. For details, see the "Transmission Configuration Model" section in *IPv4 Transmission*. In IPv6 networking, only the new transmission configuration model is supported. For details, see the "Transmission Configuration Model" section in *IPv6 Transmission*.

NOTE

You are advised to run the **SET IPGUARD** command to set the **IPGUARD.INVALIDPKCHKSW (LTE eNodeB, 5G gNodeB)** parameter to **ENABLE** so that the system reports an alarm when the number of invalid packets exceeds a specified threshold.

ACLs

An ACL is specified by **PACKETFILTER.ACLID (5G gNodeB, LTE eNodeB)** (old model)/**PACKETFILTERING.ACLID (5G gNodeB, LTE eNodeB)** (new model). An ACL consists of a set of ACL rules.

In IPv4 networking, an ACL is configured using the **ADD ACL** command; an ACL rule is configured using the **ADD ACLRULE** command.

- The **ACLRULE.VLANIDOP (5G gNodeB, LTE eNodeB)** parameter controls whether a base station filters Layer 2 packets by VLAN tag.
 - If **ACLRULE.VLANIDOP (5G gNodeB, LTE eNodeB)** is set to a value other than **OP_NOVLAN**, packets are filtered by VLAN tag. In this case, **ACLRULE.VLANID1 (5G gNodeB, LTE eNodeB)** or **ACLRULE.VLANID2 (5G gNodeB, LTE eNodeB)** must be configured.
 - If **ACLRULE.VLANIDOP (5G gNodeB, LTE eNodeB)** is set to **OP_NOVLAN**, packets without VLAN tags are filtered out.
- The base station filters Layer 3 and Layer 4 packets by combinations of the protocol type, source IP address/wildcard of the source IP address, destination IP address/wildcard of the destination IP address, source port number, destination port number, and differentiated services code point (DSCP).
 - Protocol type: specified by **ACLRULE.PT (5G gNodeB, LTE eNodeB)**
 - Source IP address: specified by **ACLRULE.SIP (5G gNodeB, LTE eNodeB)**
 - Wildcard of the source IP address: specified by **ACLRULE.SWC (5G gNodeB, LTE eNodeB)**
 - Destination IP address: specified by **ACLRULE.DIP (5G gNodeB, LTE eNodeB)**
 - Wildcard of the destination IP address: specified by **ACLRULE.DWC (5G gNodeB, LTE eNodeB)**
 - Source port number: specified by **ACLRULE.SPT1 (5G gNodeB, LTE eNodeB)**
 - Destination port number: specified by **ACLRULE.DPT1 (5G gNodeB, LTE eNodeB)**
 - DSCP: specified by **ACLRULE.DSCP (5G gNodeB, LTE eNodeB)**

In IPv6 networking, an ACL is configured using the **ADD ACL6** command; an ACL rule is configured using the **ADD ACLRULE6** command.

- Layer 2 packet filtering is the same as that in the IPv4 scenario.
- The base station filters Layer 3 and Layer 4 packets by combinations of the protocol type, source IP address/source address prefix length, destination IP address/destination address prefix length, source port number, destination port number, and differentiated services code point (DSCP).
 - Protocol type: specified by **ACLRULE6.PT (5G gNodeB, LTE eNodeB)**
 - Source IPv6 address: specified by **ACLRULE6.SIP (5G gNodeB, LTE eNodeB)**
 - Source address prefix length: specified by **ACLRULE6.SPFXLEN (5G gNodeB, LTE eNodeB)**
 - Destination IPv6 address: specified by **ACLRULE6.DIP (5G gNodeB, LTE eNodeB)**
 - Destination address prefix length: specified by **ACLRULE6.DPFXLEN (5G gNodeB, LTE eNodeB)**
 - Source port number: specified by **ACLRULE6.SPT1 (5G gNodeB, LTE eNodeB)**

- Destination port number: specified by **ACLRULE6.DPT1** (*5G gNodeB, LTE eNodeB*)
- DSCP: specified by **ACLRULE6.DSCP** (*5G gNodeB, LTE eNodeB*)

NOTE

When the NAT traversal function is enabled and ACL packet filtering is manually configured, ACL rules need to be added to allow packets whose protocol type is UDP and source and destination port numbers are both 4500 to pass through. Otherwise, packets have to undergo two filtering processes, which may affect network performance.

The IPv6 prefix length is not a wildcard mask. When the prefix length of a source address or destination address is set to **0**, the source or destination IP address can match any address.

In IPv6 networking, path maximum transmission unit (PMTU) detection prior to ACL-based packet filtering is recommended. If PMTU detection is not performed or performed after ACL-based packet filtering and ICMPv6 Packet Too Big messages returned from the intermediate network do not match any ACL rule, the PMTU of the network cannot be detected.

On an IPv6-based cascading network, the upper-level base station matches only the network layer information (source IP address, destination IP address, and DSCP), but not the transport layer information, for fragmented bypass IPv6 packets during packet filtering and IPsec ACL rule check.

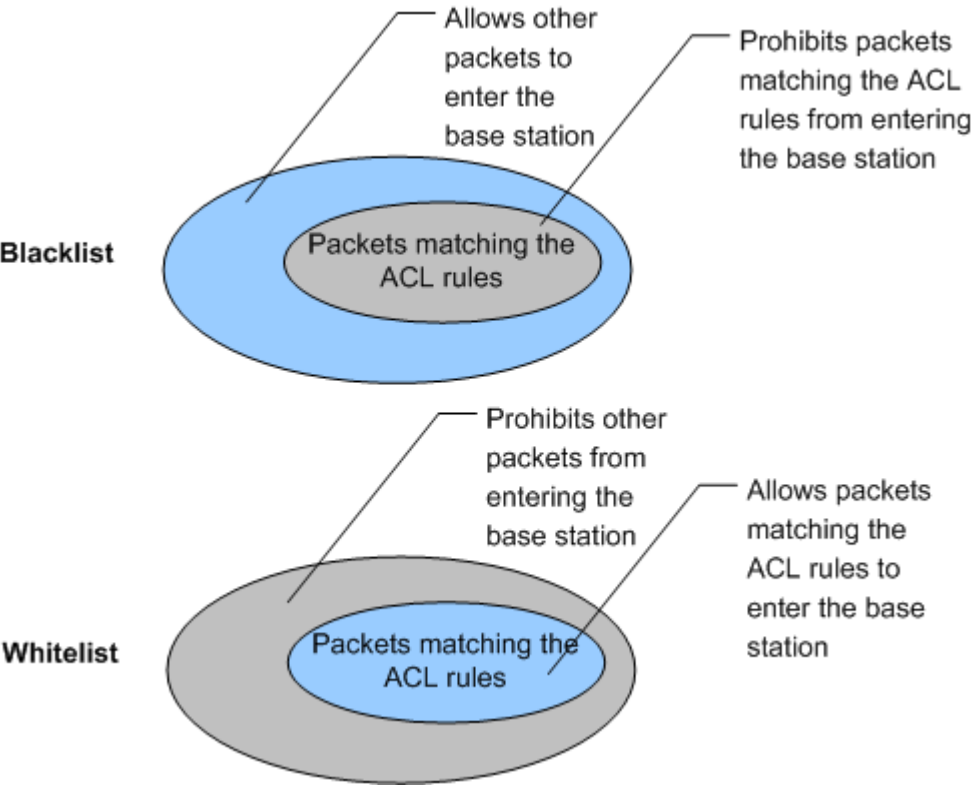
It is recommended that ACL-based packet filtering be enabled during base station running to ensure the following flows can pass ACL-based packet filtering: service-related data flows, such as OM, FTP/FTPS, IP clock link, NTP client, IPsec, DNS, PKI, S1, NG, X2, Xn, neighbor discovery (ND), and BFD data flows, and OM data flows triggered by maintenance operations, such as ping, path ping, trace route, TWAMP, IP PM, DHCP relay, UDP echo, and ICMPv6 PTB operations. Otherwise, the base station may be attacked by invalid packets. In IPsec redundancy scenarios, packet filtering rules must be considered for both the primary and secondary IPsec tunnels. For example, in IPv6 IPsec redundancy scenarios, packet filtering rules must be configured on the tunnel interfaces of both the primary IPsec tunnel and secondary IPsec tunnel.

In the new transmission configuration model, ACL-based packet filtering can be configured on ports for which the port type is set to **TUNNEL**, **INTERFACE**, **ETH**, **ETHTRK**, or **ETHCI**. The value **TUNNEL** will be removed in later versions. (In the current version, the value can still be synchronized and delivered through the interface, and the value can still take effect.) Therefore, avoid using this value. It is recommended that the value **INTERFACE** be used to indicate the backplane tunnel.

Filtering Actions

Filtering actions on a transmission port of a base station involve blacklist and whitelist configurations, as shown in [Figure 4-1](#).

Figure 4-1 Filtering action configurations on a transmission port of a base station



In IPv4 networking, the **ACLRULE.ACTION (5G gNodeB, LTE eNodeB)** parameter specifies the filtering action for packets matching the filtering rule. In IPv6 networking, the **ACLRULE6.ACTION (5G gNodeB, LTE eNodeB)** parameter specifies the filtering action for packets matching the filtering rule. The **PACKETFILTER.MB (5G gNodeB, LTE eNodeB)** (old model)/**PACKETFILTERING.MB (5G gNodeB, LTE eNodeB)** (new model) parameter specifies the filtering action for packets that do not match the filtering rule. [Table 4-1](#) provides the parameter settings of blacklist and whitelist.

NOTE

If an ACL is bound to different ports, the configurations of **PACKETFILTER.MB (5G gNodeB, LTE eNodeB)** (old model)/**PACKETFILTERING.MB (5G gNodeB, LTE eNodeB)** (new model) on these ports must be the same.

Table 4-1 Parameter settings of blacklist and whitelist

Configuration Mode	ACLRULE.ACTION (5G gNodeB, LTE eNodeB)/ ACLRULE6.ACTION (5G gNodeB, LTE eNodeB)	PACKETFILTER.MB (5G gNodeB, LTE eNodeB) (old model)/ PACKETFILTERING.MB (5G gNodeB, LTE eNodeB) (new model)
Blacklist	DENY	PERMIT
Whitelist	PERMIT	DENY

It is recommended that the whitelist mechanism be used. This is because the whitelist mechanism only requires addition of service-related data flows for ACL-based packet filtering. Compared with the blacklist mechanism, the whitelist mechanism is easier to understand and can accurately match service rules.

Precise Mistake Proofing Mechanism of ACL Rules

When automatic ACL rule configuration is disabled, the base station/BSC6960 automatically generates OMCH-related ACL rules (rule ID: 0) to prevent OMCH interruption due to incorrect manual configurations. The following lists the OMCH-related ACL rules.

- OMCH-related rule for port 6007
- IPsec rule for OMCH guarantee
- ND rule of the OMCH transmission interface in IPv6 networking scenarios

NOTE

When automatic ACL rule configuration is disabled on an interface, the precise mistake proofing mechanism of ACL rules automatically takes effect. When automatic ACL rule configuration is enabled, the OMCH is protected by automatically configured ACL rules and the precise mistake proofing mechanism does not take effect.

The ACL rules generated by the precise mistake proofing mechanism may be the same as manually configured rules. The manually configured rules may become invalid when other incorrect rules with a higher priority are configured. In contrast, the precise ACL rules have a higher priority than any other manually configured rules. This prevents the precise ACL rules from being invalid, and further prevents OMCH interruption due to incorrect manual configurations.

The precise mistake proofing mechanism of ACL rules prevents OMCH interruption only in best effort mode. Users still need to properly plan all ACL rules. This mechanism cannot provide guarantee in the following scenarios:

- The OMCH uses different interfaces for uplink and downlink transmission.
- The OMCH uses equal-cost routes.

4.1.1.2 Automatic ACL Rule Configuration

Base stations support automatic ACL rule configuration, improving whitelist configurations for ACL-based packet filtering. The automatic configuration addresses the difficulty in configuring ACL rules in whitelist mode based on the communication matrix of base stations and eliminates possible manual configuration errors.

ACL rules to be configured in whitelist mode are divided into the following two groups:

- Automatically configured ACL rule group. This group is used to ensure setups of basic services. Based on related MOs, base stations automatically configure ACL rules applicable to signaling packets, service packets, OM packets (including packets over FTP connections), clock packets, and security packets in secure networking scenarios. The ACL rules for packets over FTP connections are triggered based on the configurations of related MOs or maintenance commands for establishing FTP connections.
- Manually configured ACL rule group. For details, see [4.1.1.2.2 Restrictions on Application Scenarios](#).

In link mode, the **PACKETFILTER.ACLAUTOSWITCH (5G gNodeB, LTE eNodeB)** (old model)/**PACKETFILTERING.ACLAUTOSWITCH (5G gNodeB, LTE eNodeB)** (new model) parameter specifies whether to enable automatic ACL rule configuration.

In endpoint mode, the **EPGROUP.PACKETFILTERSWITCH (5G gNodeB, LTE eNodeB)** or **EPGROUP.PACKETFILTERSWITCH6 (LTE eNodeB, 5G gNodeB)** parameter specifies whether to enable automatic ACL rule configuration for endpoint-related MOs.

After automatic ACL rule configuration is enabled, automatically configured ACL rules cannot be manually changed, but vary with the changes in the configured MOs.

IPv4 or IPv6 ACL rules automatically configured for packet filtering are not recorded in the configuration database and can be queried using the **DSP ACLRULE** or **DSP ACLRULE6** command.

The number of ACL rules that can be configured depends on board configurations. For details, see the help of **ADD ACLRULE** and **ADD ACLRULE6**.

4.1.1.2.1 Automatic Configuration Mechanism

With automatic ACL rule configuration, a base station obtains its IP addresses and its peer NE's IP addresses based on the corresponding MO configured on the base station and then creates ACL rules for packets sent from the peer NE.

The base station checks the settings of this MO for specific packets, without considering whether the MO is being used or functional or considering the configurations of related MOs.

For example:

- For OM packets from the MAE, the base station automatically creates ACL rules based on the **OMCH** MO. If two **OMCH** MOs are configured in active/standby mode, the base station creates ACL rules for both MOs, regardless of whether the active or standby OM channel is effective.
- For signaling packets from the base station controller or MME, the base station automatically creates ACL rules based on the **SCTPLNK** MO, even if the signaling link setup fails due to an incorrect configuration or a negotiation failure.
- For security packets from the SeGW, the base station automatically creates all ACL rules based on the **IKEPEER** MO, regardless of whether this MO is referenced by the **IPSECPOLICY** or **IPSECBIND** (old model)/**IPSECBINDITF** (new model) MO.

Before automatic ACL rule configuration is enabled, the **OMCH** MO must be configured. An automatically configured ACL rule for an MO is removed or modified when the MO is removed or modified. The automatic configuration, modification, and removal of ACL rules take a period of time. If other configuration commands are run in this period, a message is normally displayed, indicating that a configuration is being exported.

NOTE

It is recommended that automatic ACL rule configuration be used to simplify ACL rule configuration and avoid configuration errors.

4.1.1.2.2 Restrictions on Application Scenarios

In the following scenarios, ACL rules must be manually configured:

- For a separate-MPT multimode base station enabled with co-transmission and cascaded base stations, ACL rules must be manually configured for passerby data flows.
- If a base station or its peer NE (such as a base station controller, MME, and S-GW) needs to be manually enabled with maintenance and testing functions that are not listed in [4.1.1.2.3 Automatically Configured ACL Rule Group](#) (such as ping, path ping, trace route, TWAMP, IP PM, DHCP relay, UDP echo, and ICMPv6 PTB operations), ACL rules for related packets must be manually configured.
- The target base station cannot obtain the user-plane IP address of the X2 interface from the source base station through the data forwarding procedure. Therefore, the ACL rules for packet filtering cannot be automatically configured and the ACL rules for the user plane of the X2 interface need to be manually configured.
- Only the BookBBU5901/BookBBU5901a/UMPTg/UMPTga/UMPTba/UMPTbb supports automatic setup and deletion of IPv6 ACL rules for packet filtering. If the preceding hardware modules have not been activated on base stations, this function does not take effect when the **PACKETFILTERING.ACLAUTOSWITCH (5G gNodeB, LTE eNodeB)** parameter is set to **ON**.

4.1.1.2.3 Automatically Configured ACL Rule Group

[Table 4-2](#) describes all the ACL rules that are automatically created based on related MOs or commands. "-" indicates that no filtering is performed.

Table 4-2 Automatically configured ACL rules

Related MO	Peer NE	Source IP Address	Source Port Number	PT	Destination IP Address	Destination Port Number
OMCH MO	MAE	MAE IP address	-	TCP	OM IP address of the base station	6007
				UDP	OM IP address of the base station	45300
				TCP	OM IP address of the base station	4443

Related MO	Peer NE	Source IP Address	Source Port Number	PT	Destination IP Address	Destination Port Number
				TCP	OM IP address of the base station	443
				TCP	OM IP address of the base station	6000
				TCP	OM IP address of the base station	6006
All MOs or maintenance commands in FTP connection mode	Control plane of the MAE FTP server	Server IP address	Modified based on the configuration of the FTPCLTPORT MO	TCP	IP address of the base station	49152-65535
	User plane of the MAE FTP server (including the PORT mode and PASV mode)	Server IP address	Determined based on STARTDATA PORT (5G gNodeB, LTE eNodeB) and ENDDATA PORT (5G gNodeB, LTE eNodeB) in the FTPCLTPORT/FTPCLTPORT 6 MO	TCP	IP address of the base station	49152-65535
IPCLKLNK MO	PTP server	Server IP address	319	UDP	IP address of the base station	319
			320	UDP	IP address of the base station	320
	IP clock server using Huawei proprietary protocol	Server IP address	35001	UDP	IP address of the base station	33003

Related MO	Peer NE	Source IP Address	Source Port Number	PT	Destination IP Address	Destination Port Number
NTPC MO	NTP server	Server IP address	Modified based on the configuration of the NTPC MO	UDP	OM IP address of the base station	9051
IKEPEER MO	SeGW	IP address of the SeGW	500	UDP	IP address of the base station	500
IKEPEER MO	SeGW (NAT)	IP address of the SeGW	500	UDP	IP address of the base station	500
			4500	UDP	IP address of the base station	4500
IKEPEER MO in endpoint mode	Peer base station	IP address of the peer base station	500	UDP	IP address of the base station	500
CA MO	CA server	Server IP address	PORT	TCP	IP address of the base station	1024-65535
CRLTSK MO with Access Method set to LDAP or HTTP	CR or CRL server	Server IP address	PORT	TCP	IP address of the base station	1024-65535
SCTPLNK MO in link mode	Base station controller/ MME/peer base station	IP address of the peer NE	PORT	SCTP	IP address of the base station	PORT
SCTPHOST and SCTPPEER MOs in endpoint mode	Base station controller/ MME/AMF/ peer base station (Abis/Iub/S1/X2/eX2/NG-C/Xn)	IP address of the peer NE	PORT	SCTP	IP address of the base station	PORT

Related MO	Peer NE	Source IP Address	Source Port Number	PT	Destination IP Address	Destination Port Number
IPPATH MO in link mode	Base station controller	IP address of the peer NE	1024-65535	UDP	IP address of the base station	1024-65535
	S-GW/UPF/peer base station	IP address of the peer NE	1024-65535	UDP	IP address of the base station	2152
USERPLANE PEER and USERPLANE HOST MOs in endpoint mode	Base station controller	IP address of the peer NE	1024-65535	UDP	IP address of the base station	1024-65535
USERPLANE PEER and USERPLANE HOST MOs in endpoint mode	S-GW/UPF/peer base station (S1/X2/NG-U/Xn)	IP address of the peer NE	1024-65535	UDP	IP address of the base station	2152
USERPLANE PEER and USERPLANE HOST MOs in endpoint mode	Peer base station (eX2)	IP address of the peer NE	1024-65535	UDP	IP address of the base station	1024-65535
IPPM MO in endpoint mode	Peer base station (X2)	IP address of the peer NE	1024-65535	UDP	IP address of the base station	65020
-	DHCP server	-	67	UDP	-	68
DNSSRV MO	DNS server	Server IP address	0-65535, determined by the server configuration ^c	UDP	Client IP address	49152-65535^a 64711^b
			0-65535, determined by the server configuration	TCP	Client IP address	49152-65535
ND	Neighborin g router	IP address of the neighborin g router	-	ICMPv6	Interface IP addresses of the base station	-

Related MO	Peer NE	Source IP Address	Source Port Number	PT	Destination IP Address	Destination Port Number
-	MME/AMF/ S-GW/UPF/ peer base station (S1/X2/NG-C/NG-U/Xn)	IP address of the peer NE	-	ICMPv6	IP address of the base station	-
BFD	Peer router	IP address of the peer router	49152-65535	UDP	IP address of the base station	Single-hop: 3784 Multi-hop: 4784 or 3784 ^d
a: A port in this range is randomly selected for DNS resolution during base station deployment by PnP. b: Port 64711 is always used for DNS resolution during normal running of base stations. c: Port 53 of the server is always used for DNS resolution during base station deployment by PnP. d: If the base station needs to use a multi-hop BFD session with destination port 3784 to interwork with the peripheral network, you need to manually specify this port number.						

NOTE

During base station deployment, no ACL rules are required because packet filtering does not take effect.

When the base station is running, ACL rules are automatically configured in the event of OM channel disconnections, regardless of the setting of the ACL rule automatic setup and deletion switch. If the OM channel is disconnected, the base station attempts to restore the channel and initiates DHCP detection. To ensure successful DHCP detection:

- The base station automatically modifies the ACL rule that filters out DHCP broadcast packets. After the DHCP detection ends, the original ACL rule is automatically restored.
- If the ACL rule cannot be modified, the base station adds an ACL rule to allow DHCP broadcast packets to enter the base station. The ID of the added ACL rule is the largest unused one within the range of 65431 to 65531. To prevent frequent ACL rule updates from affecting transmission efficiency, the base station does not remove this ACL rule immediately after the DHCP detection ends. The rule is removed only after the OM channel has been successfully established and has remained active for 30 minutes.

ACL rules are automatically configured only for IP PM links automatically configured in endpoint mode. For IP PM links automatically configured in link mode or manually configured, ACL rules are not automatically configured.

Currently, automatic ACL rule configuration for OM packets is enabled on the base station side over the following ports: 6007, 45300, 4443, 443, 6000, and 6006. The six ports are enabled by default after the **OMCH** MO is configured.

- Port 6007: used for connecting the base station to the MAE for tests, MML command execution, trace management, and alarm reporting
- Port 45300: used for receiving OM channel switchover requests from the MAE
During an OM channel switchover, the base station receives an OM channel switchover request from the MAE. In the request, the destination port number

is 45300. The MAE sends an OM channel switchover request in the following scenarios:

- The base station is configured with only one OM channel. Two OM IP addresses are configured on the MAE, and a switchover is initiated between the two addresses.
- The base station is configured with two OM channels, and a switchover is initiated between the two channels.
- The base station is configured with two OM channels. After the OM channel in use is deleted, a switchover to the other OM channel is initiated on the MAE.
- Port 4443: used for SSL-type digital certificate authentication initiated by the MAE
- Port 443: used for data configuration and maintenance in secure LMT mode
- Port 6000: used for transmitting maintenance commands and responses (in MML format) between the network information collector (NIC) and a base station
- Port 6006: used for transmitting maintenance commands and responses (in .bin format) between the NIC and a base station

When the OM channel peer IP limit switch specified by **PACKETFILTER.OMPEERIPLIMITSW (5G gNodeB, LTE eNodeB)** (old model)/**PACKETFILTERING.OMPEERIPLIMITSW (5G gNodeB, LTE eNodeB)** (new model) is set to **ON**, a base station automatically obtains the IP address used for logging in to the base station from the MAE during OM channel establishment, and changes the peer IP address in automatically configured OM channel ACL rules to this IP address. OM channel ACL rules may be frequently updated due to OM channel intermittence. To prevent frequent ACL rule updates from affecting transmission efficiency, the base station updates OM channel ACL rules only after the OM channel has been successfully established and has remained active for 30 minutes. The peer IP address is not limited immediately after the OM channel is disconnected. If IP addresses other than the IP address for logging in to the base station from the MAE are used to access the preceding ports through the MAE or another tool for OM services, ACL rules must be manually configured. These IP addresses include multiple IP addresses of a cluster MAE and the addresses of the NetEco (6007), TraceServer (6007), and NIC (6000, 6006) independently deployed from the MAE.

NOTE

It is recommended that the OM channel peer IP limit switch be turned on. If this switch is turned off, any IP address can be used to access the OM IP address of the base station, which may expose the base station to attacks.

If the source IP address in the **CRLTSK** MO is set to the default IP address **0.0.0.0** (IPv4) or **0:0:0:0:0:0:0:0** (IPv6), the OM IP address of the base station is used as the source IP address during communication. In this case, the OM IP address is used as the destination IP address (that is, the base station's IP address) in automatically configured ACL rules. If both active and standby OM IP addresses are configured, separate ACL rules for the **CRLTSK** MO are configured.

For the **NTPC** MO, the OM IP address of the base station is actually used during communication. In this case, the OM IP address is used as the destination IP address (that is, the base station's IP address) in automatically configured ACL

rules. If both active and standby OM IP addresses are configured, separate ACL rules for the **NTPC** MO are configured.

If the local IP address in the **IKEPEER** MO uses the default IP address 0.0.0.0, an interface IP address of the base station is used during communication. The base station can be configured with multiple interface IP addresses, and therefore 0.0.0.0 is used as the destination IP address (that is, the base station's IP address) in automatically configured ACL rules, in compliance with the setting in the **IKEPEER** MO. Therefore, it is recommended an appropriate interface IP address be specified in the **IKEPEER** MO when automatic ACL rule configuration is enabled.

The automatically configured ACL rules do not distinguish between boards or ports. If the IP address of a base station is configured as 0.0.0.0 or as a loopback address for data flows, the automatically configured ACL rules are added to the ACL groups where the ACL rule automatic setup and deletion switch is turned on for packet filtering. In other cases, the automatically configured ACL rules are added only to the ACL group referenced by packet filtering enabled for the port where the local IP address resides.

ACL-based packet filtering is configured on transmission ports. With this function, a base station filters packets from other NEs. If the base station has multiple transmission ports, data flows may use different inbound and outbound ports on the base station. Specifically, data flows are sent over port 1 in the uplink and received over port 2 in the downlink. In this case, it is recommended that the base station use a logical IP address.

4.1.1.2.4 Automatic ACL Rule Configuration for FTP/FTPS Packets

When a base station establishes an FTP/FTPS session for the first time, it obtains the FTP/FTPS server IP address from FTP/FTPS upload commands (for example, **ULD FILE**) and the range of the peer port number from the **FTPCLTPORT/FTPSCLTPORT6** MO, which are used to establish an ACL rule for FTP/FTPS packets. The FTP/FTPS ACL rule is not removed immediately after the FTP/FTPS session ends. The base station automatically removes the FTP/FTPS ACL rule if the rule is not used for the past consecutive 24 hours. When an FTP/FTPS session is established later, the base station determines whether an FTP/FTPS ACL rule exists. If such an ACL/FTPS rule exists, it will be used. Otherwise, a new ACL rule will be added.

NOTE

In IPv4 networking, FTP data connection (in active or passive mode) fails to be established if automatic ACL rule setup and deletion is enabled for packet filtering but the port number is beyond the range specified by the **FTPCLTPORT.STARTDATAPORT (5G gNodeB, LTE eNodeB)** and **FTPCLTPORT.ENDDATAPORT (5G gNodeB, LTE eNodeB)** parameters.

In IPv6 networking, FTP data connection (in active or passive mode) fails to be established if automatic ACL rule setup and deletion is enabled for packet filtering but the port number is beyond the range specified by the **FTPCLTPORT6.STARTDATAPORT (LTE eNodeB, 5G gNodeB)** and **FTPCLTPORT6.ENDDATAPORT (5G gNodeB, LTE eNodeB)** parameters.

FTPS is recommended because it has a higher security level than FTP.

4.1.1.2.5 ACL Rule ID Ranges

The IDs of ACL rules used for packet filtering range from 1 to 65535 and from 70000 to 74999. The IDs of ACL rules used for IPsec range from 80000 to 84999. [Table 4-3](#) describes the IDs and usage of ACL rules after the automatic ACL rule

configuration function is enabled. The ACL rule IDs are specified by the **ACLRULE.RULEID (LTE eNodeB, 5G gNodeB)** and **ACLRULE6.RULEID (LTE eNodeB, 5G gNodeB)** parameters.

Table 4-3 ACL rule ID division

ACL Rule ID	Usage
1-49999	Manually configured ACL rules
50000-50199	Automatically configured ACL rules for OM packets
50200-50299	Automatically configured ACL rules for IP clock and NTP packets
50300-50999	Automatically configured ACL rules for security packets
51000-51999	Automatically configured ACL rules for signaling packets
53000-53999	Automatically configured ACL rules for service packets
55000-59999	Automatically configured ACL rules, including those for BFD and ND packets
60000-65535	Manually configured ACL rules
70000-74499	Automatically configured ACL rules for service/signaling packets (including EPPATH packets, automatically generated IP PM packets, and packets associated with directly and indirectly connected IKE peers) in endpoint mode, which are used for packet filtering
75000-79999	Automatically configured ACL rules for SCTP service/signaling packets in endpoint mode
80000-89999	Automatically configured ACL rules for service/signaling packets (including EPPATH, SCTP, and automatically generated IP PM packets) in endpoint mode, which are used for IPsec
Note: The ACL rule IDs that are not listed in the table are reserved for future use.	

When filtering incoming data packets, the base station preferentially applies the ACL rules automatically configured in endpoint mode, and then applies the ACL rules for packet filtering. The ACL rules for packet filtering are arranged in ascending order of ACL rule IDs.

4.1.1.3 Network Attack Prevention

The ACL-based packet filtering function filters out only certain attack packets. Attackers may use IP or Media Access Control (MAC) address spoofing, where

attack packets appear to be authorized packets for access. Attackers may also use flood attacks, such as Address Resolution Protocol (ARP) and Internet Control Message Protocol (ICMP) flood attacks to attack the network. In addition, attackers may launch invalid packet attacks. If an NE receives invalid packets, the NE may experience exceptions during packet filtering based on ACL rules. For example, errors may occur or the NE may break down.

To address these security risks, flood attack prevention, invalid packet attack prevention, and ARP spoofing prevention are introduced. These functions are designed to protect against attack packets that can bypass ACL-based packet filtering. Without these functions, such attack packets may cause service quality deterioration or even interruption. It is recommended that both packet filtering and flood attack prevention be configured.

4.1.1.3.1 Rate Limitation on Broadcast Packets

Ethernet interface boards support rate limitation on broadcast packets by monitoring the number of received broadcast packets in real time to resist network storms. An alarm is reported if the broadcast packet traffic exceeds a threshold. This function is always enabled and is not configurable.

This function works as follows:

- If the number of broadcast packets received over a port per second is greater than or equal to the value of **ETHPORT.RXBCPKTALMOCRTHD (5G gNodeB, LTE eNodeB)** for 30 consecutive seconds, ALM-25879 Ethernet Port Broadcast Packets Exceeding Alarm is reported.
- If the number of broadcast packets received over a port per second is less than the value of **ETHPORT.RXBCPKTALMCLRTHD (5G gNodeB, LTE eNodeB)** for 30 consecutive seconds, this alarm is cleared.

4.1.1.3.2 ICMP Flood Attack Prevention

The following settings enable ICMP flood attack prevention and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE (5G gNodeB, LTE eNodeB)** set to **ICMP**
- **FLOODDEFEND.DFDSW (5G gNodeB, LTE eNodeB)** set to **ENABLE**
- **FLOODDEFEND.ALMSW (5G gNodeB, LTE eNodeB)** set to **ENABLE**

NOTE

It is recommended that ICMP flood attack prevention and alarm reporting be enabled. If ICMP flood attack prevention is disabled, the base station cannot defend against ICMP flood attacks. If the ICMP flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the base station encounters an ICMP flood attack.

The base station performs detection every ten seconds:

- If the number of ICMP flood packets received per second is greater than or equal to the value of **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)**, the packets are discarded and ALM-25950 Base Station Being Attacked is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** for five consecutive minutes, this alarm is cleared.

4.1.1.3.3 ICMP Response Attack Prevention

To prevent attacks on the network, the **TRANSFUNCTIONSW.ICMPPORTUNREACHABLESW** parameter can be set to **DISABLE** on a base station to disable the processing of ICMP response messages.

When the network is normal, a base station can correctly send and receive ICMP messages. However, when the network traffic is heavy, if hosts or ports are frequently unreachable, routing devices will send and receive a large number of ICMP messages, which increases the traffic load on the network and significantly reduces the performance of the routing devices. In addition, attackers often use ICMP error messages to probe the internal structure of the network.

To improve network performance and enhance network security, run the **SET TRANSFUNCTIONSW** command to disable the sending of ICMP destination unreachable messages to prevent attacks related to such ICMP messages. In addition, ICMP Redirect, ICMP Timestamp Reply, and ICMP Address Mask Reply messages are not supported. This prevents attacks related to these types of ICMP messages.

NOTE

It is recommended that the sending of ICMP response messages be disabled. If the sending of ICMP response messages is enabled for a long time, the network topology of the base station may be obtained by attackers.

4.1.1.3.4 ARP Flood Attack Prevention

Ethernet interface boards may be subject to ARP flood attacks, where attackers send a large volume of forged ARP packets with randomized source IP addresses, potentially disrupting normal communication.

The following settings enable ARP flood attack prevention and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE (5G gNodeB, LTE eNodeB)** set to **ARP**
- **FLOODDEFEND.DFDSW (5G gNodeB, LTE eNodeB)** set to **ENABLE**
- **FLOODDEFEND.ALMSW (5G gNodeB, LTE eNodeB)** set to **ENABLE**

NOTE

It is recommended that ARP flood attack prevention and alarm reporting be enabled. If ARP flood attack prevention is disabled, the base station cannot defend against ARP flood attacks. If the ARP flood alarm reporting function is disabled, the administrator will not be notified promptly when the base station encounters such attacks.

To ensure network reliability, layer 3 routing networking is recommended for connecting to peer devices. If layer 2 direct connection networking is adopted and strict ARP learning (controlled by the **IPGUARD.ARPLRNSTRICTSW (5G gNodeB, LTE eNodeB)** parameter) is not enabled, base stations may be exposed to ARP flood attacks, potentially resulting in service interruptions.

The base station performs detection every ten seconds:

- If the number of ARP flood packets received per second meets or exceeds the threshold specified by **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)**, excess packets are discarded and ALM-25950 Base Station Being Attacked is reported.

- Once this alarm is generated, it is cleared if the number of flood packets received per second remains below the threshold for five consecutive minutes.

4.1.1.3.5 ICMPv6 Flood Attack Prevention

Ethernet interface boards may be subject to ICMPv6 flood attacks, where attackers send a large volume of forged ICMPv6 packets with randomized source IP addresses, potentially disrupting normal communication. ICMPv6 has multiple sub-protocols, including ICMPv6, ND, and PTB.

The following settings enable ND flood attack prevention and alarm reporting:

- **FLOODDEFEND.FLDTYPE (5G gNodeB, LTE eNodeB)** set to **ICMP6**
- **FLOODDEFEND.FLDTYPE (5G gNodeB, LTE eNodeB)** set to **ND**
- **FLOODDEFEND.FLDTYPE (5G gNodeB, LTE eNodeB)** set to **PTB**
- **FLOODDEFEND.DFDSW (5G gNodeB, LTE eNodeB)** set to **ENABLE**
- **FLOODDEFEND.ALMSW (5G gNodeB, LTE eNodeB)** set to **ENABLE**

NOTE

It is recommended that flood attack prevention and alarm reporting functions be enabled for all the ICMPv6, ND, and PTB packets. If flood attack prevention is disabled for ICMPv6, ND, and PTB packets, the base station cannot defend against ICMPv6, ND, and PTB flood attacks. If the associated alarm reporting functions are disabled, the administrator will not be notified promptly when the base station encounters such attacks.

To ensure network reliability, layer 3 routing networking is recommended for connecting to peer devices. If layer 2 direct connection networking is adopted, base stations may be exposed to ND flood attacks, potentially resulting in service interruptions.

The base station performs detection every ten seconds:

- If the number of ICMPv6 flood packets received per second meets or exceeds the threshold specified by **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)**, excess packets are discarded and ALM-25950 Base Station Being Attacked is reported.
- Once this alarm is generated, it is cleared if the number of flood packets received per second remains below the threshold for five consecutive minutes.

4.1.1.3.6 TCP Flood Attack Prevention

The following settings enable TCP and SYN flood attacks prevention and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE (5G gNodeB, LTE eNodeB)** set to **TCP**
- **FLOODDEFEND.FLDTYPE (5G gNodeB, LTE eNodeB)** set to **SYN**
- **FLOODDEFEND.DFDSW (5G gNodeB, LTE eNodeB)** set to **ENABLE**
- **FLOODDEFEND.ALMSW (5G gNodeB, LTE eNodeB)** set to **ENABLE**

NOTE

It is recommended that TCP flood attack prevention, SYN flood attack prevention, and the corresponding alarm reporting functions be enabled. If TCP flood attack prevention and SYN flood attack prevention are disabled, the base station cannot defend against such flood attacks. If the associated alarm reporting functions are disabled, the administrator cannot receive an alarm in a timely manner when the base station encounters such flood attacks.

The base station performs detection every ten seconds:

- If the number of LACP flood packets received per second is greater than or equal to the value of the **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** parameter, the packets are discarded and ALM-25950 Base Station Being Attacked is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** for five consecutive minutes, this alarm is cleared.

4.1.1.3.7 UDP Flood Attack Prevention

The following configurations enable UDP flood attack prevention and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE (5G gNodeB, LTE eNodeB)** set to **UDP**
- **FLOODDEFEND.DFDSW (5G gNodeB, LTE eNodeB)** set to **ENABLE**
- **FLOODDEFEND.ALMSW (5G gNodeB, LTE eNodeB)** set to **ENABLE**

NOTE

It is recommended that UDP flood attack prevention and the corresponding alarm reporting function be enabled. If UDP flood attack prevention is disabled, the base station cannot defend against such flood attacks. If the associated alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the base station encounters such flood attacks.

The base station performs detection every ten seconds:

- If the number of UDP flood packets received per second is greater than or equal to the value of the **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** parameter, the packets are discarded and ALM-25950 Base Station Being Attacked is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** for five consecutive minutes, this alarm is cleared.

4.1.1.3.8 ARP/ND Spoofing Prevention

Principles for preventing ARP/ND spoofing are as follows:

- **ARP/ND entry creation:** When an interface board creates an ARP/ND entry, the entry is regarded as untrustworthy by default. If an ARP/ND entry is not updated by ARP/ND packets sent from new MAC addresses within one minute, the entry is regarded as credible and is added to the whitelist.
- **Blacklist confirmation:** An Ethernet interface board periodically checks the operations that attempt to update entries in the whitelist.
 - a. If an ARP/ND packet is received from the peer end and the packet attempts to update the MAC address in a whitelisted entry, ARP/ND requests are broadcast five times at 1-second intervals.
 - b. The sources of the received ARP/ND response packets and the number of received ARP/ND response packets are determined.

If...	Then...
Three or more ARP/ND response packets are received from a MAC address that is in a whitelisted ARP/ND entry	The MAC address is considered as credible.
Three or more ARP/ND response packets are received from a MAC address that is not in any whitelisted ARP/ND entry	The MAC address is added to the blacklist.

The base station's integrated IP protocol stack processing unit starts ARP/ND spoofing detection when receiving ARP/ND packets that attempt to update an ARP/ND entry. If the detection result indicates that the original ARP/ND entry is credible, the received ARP/ND packets are regarded as spoofed ARP/ND packets. The base station then adds the MAC address of such packets to a blacklist and does not use this MAC address to generate ARP/ND entries before the blacklist expires.

ARP spoofing prevention is enabled for a base station when the **IPGUARD.ARPSPOOFCHKSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ENABLE**. If the number of discarded spoofed ARP packets is greater than or equal to the value of the **IPGUARD.ARPSPOOFALMTHD (LTE eNodeB, 5G gNodeB)** parameter after ARP spoofing prevention is enabled, the base station reports ALM-25950 Base Station Being Attacked. Information about the discarded spoofed ARP packets can be queried using the **DSP INVALIDPKTINFO** command.

ND spoofing prevention is enabled for a base station when the **IPGUARD.NDSPOOFCHKSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ENABLE**. If the number of discarded spoofed ND packets is greater than or equal to the value of the **IPGUARD.NDSPOOFALMTHD (LTE eNodeB, 5G gNodeB)** parameter after ND spoofing prevention is enabled, the base station reports ALM-25950 Base Station Being Attacked. Information about the discarded spoofed ND packets can be queried using the **DSP INVALIDPKTINFO** command.

NOTE

- It is recommended that ARP spoofing prevention and ND spoofing prevention be enabled. If the ARP spoofing prevention function is disabled, the base station cannot defend against ARP spoofing attacks. If the ND spoofing prevention function is not enabled, the base station cannot defend against ND spoofing attacks.
- If the MAC address of a network device is added to the ARP/ND blacklist, the base station still responds to the ARP/ND request initiated by the device. In this way, the peer device can learn the MAC address of the base station.
- If the OMCH is disconnected and the MAC address in the ARP entry used when the OMCH is normal is mistakenly added to the blacklist, the base station performs ARP blacklist confirmation again.

The **IPGUARD.ARPLRNSTRICTSW (5G gNodeB, LTE eNodeB)** parameter specifies whether to enable strict ARP learning. In the learning without ARP entries, an NE learns only the ARP response packets in response to the ARP request packets sent by itself. This function allows a base station to deny most attacks from ARP requests and responses.

- If this parameter is set to **ENABLE**, strict ARP learning is enabled. After an ARP entry is generated, you can configure its update policy using the **IPGUARD.ARPLRNSTRICTPOLICY (LTE eNodeB, 5G gNodeB)** parameter.
 - If this parameter is set to **ALL**, the NE updates entries after receiving ARP request and response packets.
 - If this parameter is set to **ONLY_RESPONSE**, the NE does not update ARP entries after receiving ARP request packets but updates ARP entries after receiving ARP response packets.

 NOTE

IPGUARD.ARPLRNSTRICTPOLICY (LTE eNodeB, 5G gNodeB) can be configured and take effect only when **IPGUARD.ARPLRNSTRICTSW (5G gNodeB, LTE eNodeB)** is set to **ENABLE**.

- If this parameter is set to **DISABLE**, strict ARP learning will be disabled. In this case, the ARP entries of the NE may be exhausted by attackers.

4.1.1.3.9 IPv6 SEND

The ND protocol plays an important role in the IPv6 protocol suite. As network security problems intensify, RFC3971 SEcure Neighbor Discovery (SEND) extends the ND protocol to improve security. Base stations support the following extension options:

- Timestamp option: It is used to ensure that unsolicited notification messages and redirection messages are not replayed.
- Nonce option: It is used to ensure that a notification message is the first response to a previous request message sent by the node.

Timestamp and Nonce options are used to prevent replay attacks. CGA and RSA options are not supported.

4.1.1.3.10 Smurf Attack Prevention

Ethernet interface boards of the base station support Smurf attack prevention, thereby preventing network congestion due to Smurf attacks. The Smurf attack prevention function is always enabled and is not configurable.

The Smurf attack prevention function checks the received ICMP packet as follows:

- If the destination address of the packet is a network or broadcast address, the packet is discarded.
- If the destination address of the packet is the address of the interface board, the packet is allowed to pass through.

4.1.1.3.11 Malformed Packet Attack Prevention

To prevent malformed packet attacks, a base station checks whether received packets have the characteristics of malformed packets.

The types of malformed packets that can be prevented include TCP local area network denial (LAND) and IP packets with incorrect field values. A base station directly discards such packets after identification. Currently, certain information about discarded packets can be queried by running the **DSP INVALIDPKTINFO** command with **DRPPKTTYPE** set to **MALFORM**. The information includes

incorrect IP version number, incorrect IPv4 header length, all-0 IPv4 destination IP address, broadcast IPv4 source IP address, and incorrect IPv6 header length.

Malformed packet prevention starts automatically after base station startup and cannot be configured.

4.1.1.3.12 SCTP Flood Attack Prevention

Interface boards may experience SCTP flood attacks in which attackers send a large number of spoofed SCTP packets to interface boards, interrupting the communication.

The following settings enable SCTP flood attack prevention and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE** (*5G gNodeB, LTE eNodeB*) set to **SCTP**
- **FLOODDEFEND.DFDSW** (*5G gNodeB, LTE eNodeB*) set to **ENABLE**
- **FLOODDEFEND.ALMSW** (*5G gNodeB, LTE eNodeB*) set to **ENABLE**

NOTE

It is recommended that SCTP flood attack prevention and alarm reporting be enabled. If SCTP flood attack prevention is disabled, the base station cannot defend against SCTP flood attacks. If the SCTP flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the base station encounters an SCTP flood attack.

The base station performs detection every ten seconds:

- If the number of SCTP flood packets received per second is greater than or equal to the value of **FLOODDEFEND.DFDTHD** (*5G gNodeB, LTE eNodeB*), the packets are discarded and ALM-25950 Base Station Being Attacked is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of **FLOODDEFEND.DFDTHD** (*5G gNodeB, LTE eNodeB*) for five consecutive minutes, this alarm is cleared.

4.1.1.3.13 IKE Flood Attack Prevention

The following settings enable IKE flood attack prevention and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE** (*LTE eNodeB, 5G gNodeB*) set to **IKE**
- **FLOODDEFEND.DFDSW** (*LTE eNodeB, 5G gNodeB*) set to **ENABLE**
- **FLOODDEFEND.ALMSW** (*LTE eNodeB, 5G gNodeB*) set to **ENABLE**

NOTE

It is recommended that IKE flood attack prevention and alarm reporting be enabled. If IKE flood attack prevention is disabled, the base station cannot defend against IKE flood attacks. If the IKE flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the base station encounters an IKE flood attack.

The base station performs detection every ten seconds:

- If the number of IKE flood packets received per second is greater than or equal to the value of the **FLOODDEFEND.DFDTHD** (*LTE eNodeB, 5G gNodeB*) parameter, the packets are discarded and ALM-25950 Base Station Being Attacked is reported.

- After this alarm is generated, if the number of flood packets received per second is less than the value of the **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** parameter for five consecutive minutes, this alarm is cleared.

4.1.1.3.14 802.1x Flood Attack Prevention

The following settings enable 802.1x flood attack prevention and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE (LTE eNodeB, 5G gNodeB)** set to **DOT1X**
- **FLOODDEFEND.DFDSW (LTE eNodeB, 5G gNodeB)** set to **ENABLE**
- **FLOODDEFEND.ALMSW (LTE eNodeB, 5G gNodeB)** set to **ENABLE**

NOTE

It is recommended that 802.1x flood attack prevention and alarm reporting be enabled. If 802.1x flood attack prevention is disabled, the base station cannot defend against 802.1x flood attacks. If the 802.1x flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the base station encounters an 802.1x flood attack.

The base station performs detection every ten seconds:

- If the number of 802.1x flood packets received per second is greater than or equal to the value of the **FLOODDEFEND.DFDTHD (LTE eNodeB, 5G gNodeB)** parameter, the packets are discarded and ALM-25950 Base Station Being Attacked is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of the **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** parameter for five consecutive minutes, this alarm is cleared.

4.1.1.3.15 LACP Flood Attack Prevention

The following settings enable LACP flood attack prevention and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE (LTE eNodeB, 5G gNodeB)** set to **LACP**
- **FLOODDEFEND.DFDSW (LTE eNodeB, 5G gNodeB)** set to **ENABLE**
- **FLOODDEFEND.ALMSW (LTE eNodeB, 5G gNodeB)** set to **ENABLE**

NOTE

It is recommended that LACP flood attack prevention and alarm reporting be enabled. If LACP flood attack prevention is disabled, the base station cannot defend against LACP flood attacks. If the LACP flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the base station encounters an LACP flood attack.

The base station performs detection every ten seconds:

- If the number of LACP flood packets received per second is greater than or equal to the value of the **FLOODDEFEND.DFDTHD (LTE eNodeB, 5G gNodeB)** parameter, the packets are discarded and ALM-25950 Base Station Being Attacked is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of the **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** parameter for five consecutive minutes, this alarm is cleared.

4.1.1.3.16 GTP-U Error Indication Flood Attack Defense

The base station classifies GTP-U packets to be sent for control-plane processing into two types: GTP-U Error Indication packets and GTP-U control packets, and provides different flood attack prevention thresholds.

The following settings enable GTP-U Error Indication flood attack prevention and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE (LTE eNodeB, 5G gNodeB)** set to **GTPU_ERR_IND**
- **FLOODDEFEND.DFDSW (LTE eNodeB, 5G gNodeB)** set to **ENABLE**
- **FLOODDEFEND.ALMSW (LTE eNodeB, 5G gNodeB)** set to **ENABLE**

NOTE

It is recommended that GTP-U Error Indication flood attack prevention and alarm reporting be enabled. If GTP-U Error Indication flood attack prevention is disabled, the base station cannot defend against GTP-U Error Indication flood attacks. If the GTP-U Error Indication flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the base station encounters a GTP-U Error Indication flood attack.

The base station performs detection every ten seconds:

- If the number of GTP-U Error Indication flood packets received per second is greater than or equal to the value of the **FLOODDEFEND.DFDTHD (LTE eNodeB, 5G gNodeB)** parameter, the packets are discarded and ALM-25950 Base Station Being Attacked is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of the **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** parameter for five consecutive minutes, this alarm is cleared.

4.1.1.3.17 GTP-U Control Flood Attack Defense

The base station classifies GTP-U packets to be sent for control-plane processing into two types: GTP-U Error Indication packets and GTP-U control packets, and provides different flood attack prevention thresholds. The types of GTP-U control packets include echo requests, echo responses, notifications for supported extension headers, and 3GPP-reserved messages.

The following settings enable GTP-U control flood attack prevention and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE (LTE eNodeB, 5G gNodeB)** set to **GTPU_CTRL**
- **FLOODDEFEND.DFDSW (LTE eNodeB, 5G gNodeB)** set to **ENABLE**
- **FLOODDEFEND.ALMSW (LTE eNodeB, 5G gNodeB)** set to **ENABLE**

NOTE

It is recommended that GTP-U control flood attack prevention and alarm reporting be enabled. If GTP-U control flood attack prevention is disabled, the base station cannot defend against GTP-U control flood attacks. If the GTP-U control flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the base station encounters a GTP-U control flood attack.

The base station performs detection every ten seconds:

- If the number of GTP-U control flood packets received per second is greater than or equal to the value of the **FLOODDEFEND.DFDTHD (LTE eNodeB, 5G**

gNodeB) parameter, the packets are discarded and ALM-25950 Base Station Being Attacked is reported.

- After this alarm is generated, if the number of flood packets received per second is less than the value of the **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** parameter for five consecutive minutes, this alarm is cleared.

4.1.1.3.18 1588 L2 Multicast Packet Flood Attack Defense

The following settings enable 1588 L2 multicast packet flood attack defense and alarm reporting on the base station side:

- **FLOODDEFEND.FLDTYPE (LTE eNodeB, 5G gNodeB)** set to **1588V2_L2_MULTICAST**
- **FLOODDEFEND.DFDSW (LTE eNodeB, 5G gNodeB)** set to **ENABLE**
- **FLOODDEFEND.ALMSW (LTE eNodeB, 5G gNodeB)** set to **ENABLE**

NOTE

It is recommended that 1588 L2 multicast packet flood attack defense and alarm reporting be enabled. If 1588 L2 multicast packet flood attack defense is disabled, the base station cannot defend against such attacks. If the 1588 L2 multicast packet flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the base station encounters such attacks.

The base station performs detection every ten seconds:

- If the number of 1588 L2 multicast flood packets received per second is greater than or equal to the value of the **FLOODDEFEND.DFDTHD (LTE eNodeB, 5G gNodeB)** parameter, the packets are discarded and ALM-25950 Base Station Being Attacked is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of the **FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)** parameter for five consecutive minutes, this alarm is cleared.

4.1.2 Integrated Firewall of the BSC6900/BSC6910 (GSM/UMTS)

4.1.2.1 ACL-based Packet Filtering

The procedure for configuring ACL rules and enabling packet filtering on the base station controller side is similar to that on the base station side. Their differences are as follows:

- The filtering action does not need to be configured and only the whitelist function is supported on the base station controller side.
- On the base station controller side, only filtering rules based on the destination IP address specified by the **ACLRULE.DIP** parameter are supported.

In addition to manually configured ACL rules, the base station controller automatically generates ACLs in advance for incoming packets. This function is called the intelligent whitelist function. The filter criteria for ACL rules of the intelligent whitelist function include the source IP address, destination IP address, port number, protocol type, and DSCP priority. After receiving a packet, the base

station controller checks whether the packet matches the ACL rules. If it matches the ACL rules, the base station controller accepts the packet. If it does not match the ACL rules, the base station controller discards the packet. The intelligent whitelist function is always enabled and is not configurable.

4.1.2.2 Network Attack Prevention

4.1.2.2.1 Rate Limitation on Broadcast Packets

Ethernet interface boards support rate limitation on broadcast packets by monitoring the number of received broadcast packets in real time to resist network storms. An alarm is reported if the broadcast packet traffic exceeds a threshold. This function is always enabled and is not configurable.

This function works as follows:

- If the number of broadcast packets received over a port per second is greater than or equal to the value of **ETHPORT.BCPKTALARMTHD** for 30 consecutive seconds, ALM-21387 Ethernet Port Broadcast Packets Exceeding Alarm is reported.
- If the number of broadcast packets received over a port per second is less than the value of **ETHPORT.BCPKTALARMCLRTHD** for 30 consecutive seconds, this alarm is cleared.

4.1.2.2.2 ICMP Flood Attack Prevention

For the base station controller, you can run the **ADD ICMPGUARD** command to configure ICMP attack defense policies. With these policies, interface boards discard the specified types of ICMP packets sent from IP addresses in the specified network segment.

- The **ICMPGUARD.IPADDR** parameter specifies the source IP address of ICMP attack packets.
- The **ICMPGUARD.GUARDTYPE** parameter specifies the type of ICMP attack packets.

When the **IPGUARD.ICMPALMSW** parameter is set to **ON(ON)**, interface boards on the base station controller monitor the traffic of ICMP attack packets in real time. In 30 consecutive seconds:

- If the number of ICMP attack packets received over an interface board per second is greater than or equal to the value of **IPGUARD.ICMPALMTHD**, the base station controller discards ICMP packets and reports ALM-21388 Invalid Packets Exceeding Alarm.
- If the number of ICMP attack packets received over an interface board per second is less than the value of **IPGUARD.ICMPALMRTHD**, the base station controller clears this alarm and does not discard ICMP packets.

4.1.2.2.3 ARP Flood Attack Prevention

Interface boards may experience ARP flood attacks in which attackers send to interface boards a large number of spoofed ARP packets whose source IP addresses have been tampered with, interrupting the communication.

For the base station controller, the ARP entry learning function is used to defense against ARP flood attacks. This function is controlled by the **IPGUARD.ARPARNSTRICTSW** parameter and is enabled by default.

With this function, interface boards learn only from the ARP response packets corresponding to the ARP request packets sent from the local system and record the MAC addresses in the response packets. This enables interface boards to reject spoofed ARP packets.

4.1.2.2.4 ARP Spoofing Prevention

For the base station controller, ARP spoofing prevention is controlled by the **IPGUARD.ARPANTICHEATSW** parameter.

If an interface board detects more than 30 ARP entry update attempts (excluding those from credible MAC addresses) within 1 minute, the interface board reports ALM-21391 ARP Conflict. The alarm parameter **Attacker's MAC Address** specifies the MAC address that has the most ARP entry update attempts within the last credible-ARP-entry decision period before the alarm is reported. The source of an ARP spoofing attack can be identified in the following ways:

- After the alarm is reported, check the value of **Attacker's MAC Address** to find out the source. Run the **DSP ARSPOOFING** command to find out the sources of all ARP spoofing attacks.
- After the alarm is cleared or the blacklist is aged, check the value of **Attacker's MAC Address** in historical alarms to find out the sources of historical ARP spoofing attacks.

If the interface board does not detect any ARP entry update attempts within 1 minute after this alarm is reported, the alarm is cleared.

NOTE

ALM-21391 ARP Conflict applies to IP addresses in the ARP entries. For IP addresses that are not included in the ARP entries, for example, IP address of an interface board, ALM-21347 IP Address Conflict applies.

4.1.2.2.5 Smurf Attack Prevention

Ethernet interface boards of the base station controller support Smurf attack prevention, thereby preventing network congestion due to Smurf attacks. The Smurf attack prevention function is always enabled and is not configurable.

With this function, an interface board checks the destination IP address of each received ICMP packet:

- If the destination IP address of the packet is a network or broadcast address, the interface board discards the packet.
- If the destination IP address of the packet is the interface board's IP address, the interface board accepts the packet.

4.1.2.2.6 Illegal Packet Attack Prevention

An illegal packet may be an illegal IP packet, multicast MAC packet, or ICMP packet.

- Illegal IP packet
The packets filtered by an integrated firewall are illegal IP packets.
- Invalid multicast MAC packet
A multicast MAC packet is illegal if it is received over an interface board for which the ETH Operation, Administration, and Maintenance (ETH OAM) function is not enabled.
- Invalid ICMP packet
An invalid ICMP packet is defined by the **ICMPGUARD.GUARDTYPE** parameter.

For the base station controller, the invalid packet attack prevention function is always enabled and is not configurable. An alarm is reported when an interface board receives excessive illegal packets.

- Interface boards monitor the traffic of illegal IP packets in real time when the **IPGUARD.VALIDPKTCHKSW** parameter is set to **ON(ON)**.
 - If the number of illegal IP packets received by an interface board per second is greater than or equal to the value of **IPGUARD.INVALIDPKTALMTHD** for 30 consecutive seconds, ALM-21388 Invalid Packets Exceeding Alarm is reported.
 - If the number of illegal IP packets received by an interface board per second is less than the value of **IPGUARD.INVALIDPKTALMRTHD** for 30 consecutive seconds, this alarm is cleared.
- Interface boards monitor the traffic of illegal multicast MAC packets in real time when the **IPGUARD.INVALIDMCASTALMSW** parameter is set to **ON(ON)**.
 - If the number of illegal multicast MAC packets received by an interface board per second is greater than or equal to the value of **IPGUARD.INVALIDMCASTALMTHD** for 30 consecutive seconds, ALM-21388 Invalid Packets Exceeding Alarm is reported.
 - If the number of illegal multicast MAC packets received by an interface board per second is less than the value of **IPGUARD.INVALIDMCASTALMRTHD** for 30 consecutive seconds, this alarm is cleared.

When users are notified of the ALM-21388 Invalid Packets Exceeding Alarm, they can locate the attack source and determine the attack type based on the statistics and detailed information about illegal packets.

- Search the operation logs for statistics on illegal packets.
- Run the **DSP INVALIDPKTINFO** command to obtain detailed information about illegal packets.

4.1.3 BSC6960 Integrated Firewall

The BSC6960 deployed on networks may encounter various network attacks. Therefore, lightweight firewalls are integrated into BSC6960s to support ACL-based packet filtering, flood attack prevention, invalid packet attack prevention, and other functions.

4.1.3.1 ACL-based Packet Filtering

On the BSC6960 side, ACL rules are used to filter invalid Layer 2 (data link layer), Layer 3 (network layer), and Layer 4 (transport layer) packets. ACL-based packet filtering involves ACLs and filtering actions. For details, see section "Transmission Configuration Model" in *IPv4 Transmission*.

NOTE

You are advised to run the **SET IPGUARD** command to set the **IPGUARD.INVALIDPKTCHKSW** parameter to **ENABLE** so that the system reports an alarm when the number of invalid packets exceeds a specified threshold.

ACLs

PACKETFILTERING.ACLID specifies the ID of an ACL. An ACL consists of a set of ACL rules.

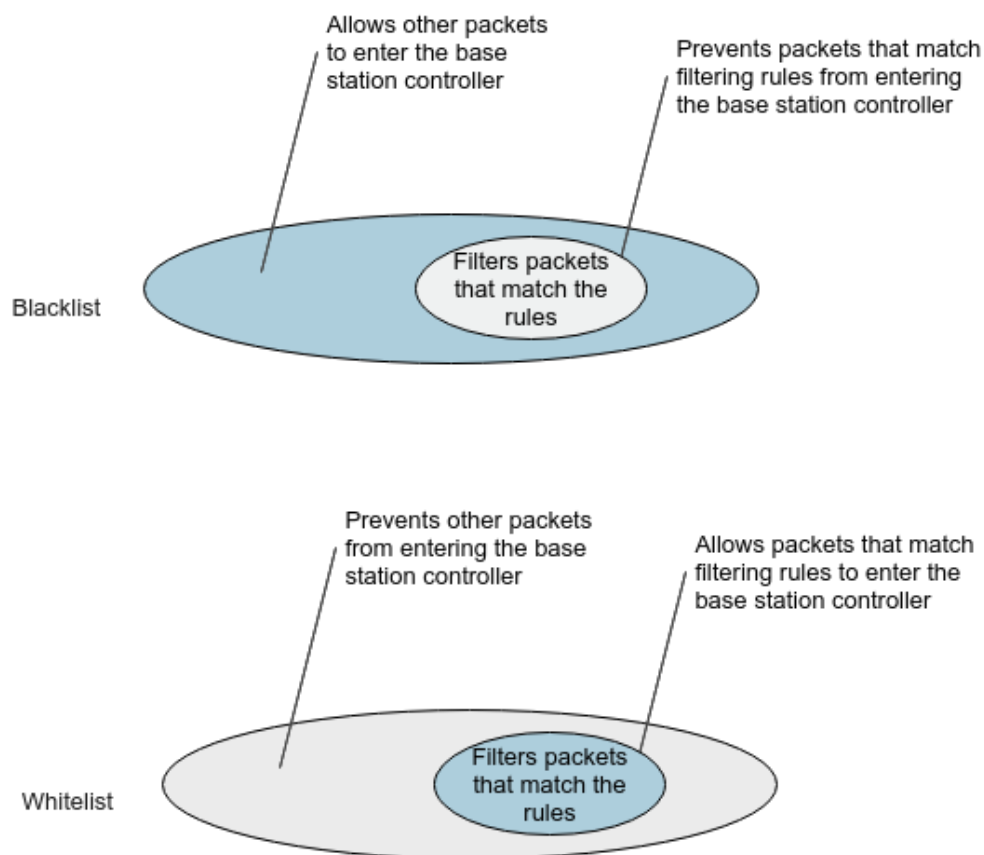
In IPv4 networking, an ACL is configured using the **ADD ACL** command; an ACL rule is configured using the **ADD ACLRULE** command.

- The **ACLRULE.VLANIDOP** parameter controls whether the BSC6960 filters Layer 2 packets by VLAN tag.
 - If **ACLRULE.VLANIDOP** is set to a value other than **OP_NOVLAN**, packets are filtered by VLAN tag. In this case, **ACLRULE.VLANID1** or **ACLRULE.VLANID2** must be configured.
 - If **ACLRULE.VLANIDOP** is set to **OP_NOVLAN**, packets without VLAN tags are filtered out.
- The BSC6960 filters Layer 3 and Layer 4 packets by combinations of the protocol type, source IP address/wildcard of the source IP address, destination IP address/wildcard of the destination IP address, source port number, destination port number, and differentiated services code point (DSCP).
 - Protocol type: specified by **ACLRULE.PT**
 - Source IP address: specified by **ACLRULE.SIP**
 - Wildcard of the source IP address: specified by **ACLRULE.SWC**
 - Destination IP address: specified by **ACLRULE.DIP**
 - Wildcard of the destination IP address: specified by **ACLRULE.DWC**
 - Source port number: specified by **ACLRULE.SPT1**
 - Destination port number: specified by **ACLRULE.DPT1**
 - DSCP: specified by **ACLRULE.DSCP**

Filtering Actions

Filtering actions on a transmission port of a BSC6960 involve blacklist and whitelist configurations, as shown in [Figure 4-2](#).

Figure 4-2 Filtering action configurations on a transmission port of a BSC



In IPv4 address scenarios, the **ACLRULE.ACTION** parameter specifies the rules for filtering packets that match ACL rules. The **PACKETFILTERING.MB** parameter specifies the rules for filtering packets that do not match ACL rules. [Table 4-4](#) provides the parameter settings of blacklist and whitelist.

NOTE

If an ACL is bound to different ports, the configurations of **PACKETFILTERING.MB** on these ports must be the same.

Table 4-4 Parameter settings of blacklist and whitelist

Configuration Mode	ACLRULE.ACTION	PACKETFILTERING.MB
Blacklist	DENY	PERMIT
Whitelist	PERMIT	DENY

It is recommended that the whitelist mechanism be used. This is because the whitelist mechanism only requires addition of service-related data flows for ACL-based packet filtering. Compared with the blacklist mechanism, the whitelist mechanism is easier to understand and can accurately match service rules.

4.1.3.2 Network Attack Prevention

The ACL-based packet filtering function filters out only certain attack packets. Attackers may use IP or Media Access Control (MAC) address spoofing, where attack packets appear to be authorized packets for access. Attackers may also use flood attacks, such as Address Resolution Protocol (ARP) and Internet Control Message Protocol (ICMP) flood attacks to attack the network. In addition, attackers may launch invalid packet attacks. If an NE receives invalid packets, the NE may experience exceptions during packet filtering based on ACL rules. For example, errors may occur or the NE may break down.

To address these security risks, flood attack prevention, invalid packet attack prevention, and ARP spoofing prevention are introduced. These functions are designed to protect against attack packets that can bypass ACL-based packet filtering. Without these functions, such attack packets may cause service quality deterioration or even interruption. It is recommended that both packet filtering and flood attack prevention be configured.

4.1.3.2.1 Rate Limitation on Broadcast Packets

Ethernet interface boards support rate limitation on broadcast packets by monitoring the number of received broadcast packets in real time to resist network storms. An alarm is reported if the broadcast packet traffic exceeds a threshold. This function is always enabled and is not configurable.

This function works as follows:

- If the number of broadcast packets received over a port per second is greater than or equal to the value of **ETHPORT.RXBCPKTALMOCRTHD** for 30 consecutive seconds, ALM-21650 Ethernet Port Broadcast Packets Exceeding Alarm is reported.
- If the number of broadcast packets received over a port per second is less than the value of **ETHPORT.RXBCPKTALMOCRTHD** for 30 consecutive seconds, this alarm is cleared.

4.1.3.2.2 ICMP Flood Attack Prevention

The following settings enable ICMP flood attack prevention and alarm reporting on the BSC6960 side:

- **FLOODDEFEND.FLDTYPE** set to **ICMP**
- **FLOODDEFEND.DFDSW** set to **ENABLE**
- **FLOODDEFEND.ALMSW** set to **ENABLE**

NOTE

It is recommended that ICMP flood attack prevention and alarm reporting be enabled. If ICMP flood attack prevention is disabled, the BSC6960 cannot defend against ICMP flood attacks. If the ICMP flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the BSC6960 encounters an ICMP flood attack.

The BSC6960 checks the following items every 10 seconds:

- If the number of ICMP flood packets received per second is greater than or equal to the value of **FLOODDEFEND.DFDTHD**, the packets are discarded and ALM-21665 Invalid Packets Exceeding Alarm is reported.

- After this alarm is generated, if the number of flood packets received per second is less than the value of **FLOODDEFEND.DFDTHD** for five consecutive minutes, this alarm is cleared.

4.1.3.2.3 ICMP Response Attack Prevention

To prevent attacks on the network, the **TRANSFUNCTIONSW.ICMPPORTUNREACHABLESW** parameter can be set to **DISABLE** on a BSC6960 to disable the processing of ICMP response messages.

When the network is normal, a base station can correctly send and receive ICMP messages. However, when the network traffic is heavy, if hosts or ports are frequently unreachable, routing devices will send and receive a large number of ICMP messages, which increases the traffic load on the network and significantly reduces the performance of the routing devices. In addition, attackers often use ICMP error messages to probe the internal structure of the network.

To improve network performance and enhance network security, run the **SET TRANSFUNCTIONSW** command to disable the sending of ICMP destination unreachable messages to prevent attacks related to such ICMP messages. In addition, ICMP Redirect, ICMP Timestamp Reply, and ICMP Address Mask Reply messages are not supported. This prevents attacks related to these types of ICMP messages.

NOTE

It is recommended that the sending of ICMP response messages be disabled. If the sending of ICMP response messages is enabled for a long time, the network topology of the BSC6960 may be obtained by attackers.

4.1.3.2.4 ARP Flood Attack Prevention

Interface boards may experience ARP flood attacks in which attackers send to interface boards a large number of spoofed ARP packets whose source IP addresses have been tampered with, interrupting the communication.

The following settings enable ARP flood attack prevention and alarm reporting on the BSC6960 side:

- **FLOODDEFEND.FLDTYPE** set to **ARP**
- **FLOODDEFEND.DFDSW** set to **ENABLE**
- **FLOODDEFEND.ALMSW** set to **ENABLE**

NOTE

It is recommended that ARP flood attack prevention and alarm reporting be enabled. If ARP flood attack prevention is disabled, the BSC6960 cannot defend against ARP flood attacks. If the ARP flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the BSC6960 encounters an ARP flood attack.

The BSC6960 checks the following items every 10 seconds:

- If the number of ARP flood packets received per second is greater than or equal to the value of **FLOODDEFEND.DFDTHD**, the packets are discarded and ALM-21665 Invalid Packets Exceeding Alarm is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of **FLOODDEFEND.DFDTHD** for five consecutive minutes, this alarm is cleared.

4.1.3.2.5 TCP Flood Attack Prevention

The following settings enable TCP and SYN flood attacks prevention and alarm reporting on the BSC6960 side:

- **FLOODDEFEND.FLDTYPE** set to **TCP**
- **FLOODDEFEND.FLDTYPE** set to **SYN**
- **FLOODDEFEND.DFDSW** set to **ENABLE**
- **FLOODDEFEND.ALMSW** set to **ENABLE**

 NOTE

It is recommended that TCP flood attack prevention, SYN flood attack prevention, and the corresponding alarm reporting functions be enabled. If TCP flood attack prevention and SYN flood attack prevention are disabled, the BSC6960 cannot defend against such flood attacks. If the associated alarm reporting functions are disabled, the administrator cannot receive an alarm in a timely manner when the BSC6960 encounters such flood attacks.

The BSC6960 checks the following items every 10 seconds:

- If the number of TCP flood packets received per second is greater than or equal to the value of **FLOODDEFEND.DFDTHD**, the packets are discarded and ALM-21665 Invalid Packets Exceeding Alarm is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of **FLOODDEFEND.DFDTHD** for five consecutive minutes, this alarm is cleared.

4.1.3.2.6 UDP Flood Attack Prevention

The following settings enable UDP flood attack prevention and alarm reporting on the BSC6960 side:

- **FLOODDEFEND.FLDTYPE** set to **UDP**
- **FLOODDEFEND.DFDSW** set to **ENABLE**
- **FLOODDEFEND.ALMSW** set to **ENABLE**

 NOTE

It is recommended that UDP flood attack prevention and the corresponding alarm reporting function be enabled. If UDP flood attack prevention is disabled, the BSC6960 cannot defend against such flood attacks. If the associated alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the BSC6960 encounters such flood attacks.

The BSC6960 checks the following items every 10 seconds:

- If the number of UDP flood packets received per second is greater than or equal to the value of **FLOODDEFEND.DFDTHD**, the packets are discarded and ALM-21665 Invalid Packets Exceeding Alarm is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of **FLOODDEFEND.DFDTHD** for five consecutive minutes, this alarm is cleared.

4.1.3.2.7 ARP Spoofing Prevention

Principles for preventing ARP spoofing are as follows:

- **ARP entry creation:** When a board creates an ARP entry, the entry is regarded as untrustworthy by default. If an ARP entry is not updated by ARP packets sent from new MAC addresses within one minute, the entry is regarded as credible and is added to the whitelist.
- **Blacklist confirmation:** An Ethernet interface board periodically checks the operations that attempt to update entries in the whitelist.
 - a. If an ARP packet is received from the peer end and the packet attempts to update the MAC address in a whitelisted entry, ARP requests are broadcast five times at 1-second intervals.
 - b. The sources of the received ARP response packets and the number of received ARP/ND response packets are determined.

If...	Then...
The interface board receives three or more ARP response packets from a MAC address that is in a whitelisted ARP entry	The MAC address is considered as credible.
The interface board receives three or more ARP response packets from a MAC address that is not in a whitelisted ARP entry	The MAC address is added to the blacklist.

After receiving the ARP packets that are to replace the original entities in the ARP list, the processing module of the IP protocol stack on the BSC6960 performs the ARP spoofing detection. If the detection result indicates that the original ARP table is credible, the received ARP packets are regarded as spoofed ARP packets. Then, the BSC6960 adds the MAC address of the ARP spoofing packets to a blacklist and does not process ARP packets containing this MAC address before the blacklist expires.

ARP spoofing prevention is enabled for a BSC6960 when the **IPGUARD.ARPSPOOFCHKSW** parameter is set to **ENABLE**. If the number of discarded spoofed ARP packets is greater than or equal to the value of the **IPGUARD.ARPSPOOFALMTHD** parameter after ARP spoofing prevention is enabled, the BSC6960 reports ALM-21665 Invalid Packets Exceeding Alarm. Information about the discarded spoofed ARP packets can be queried using the **DSP INVALIDPKTINFO** command.

NOTE

It is recommended that ARP spoofing prevention be enabled. If ARP spoofing prevention is disabled, the BSC6960 cannot defend against ARP spoofing attacks.

4.1.3.2.8 Smurf Attack Prevention

Ethernet interface boards of the BSC6960 support Smurf attack prevention, thereby preventing network congestion due to Smurf attacks. The Smurf attack prevention function is always enabled and is not configurable.

The Smurf attack prevention function checks the received ICMP packet as follows:

- If the destination address of the packet is a network or broadcast address, the packet is discarded.

- If the destination address of the packet is the address of the board, the packet is allowed to pass through.

4.1.3.2.9 Malformed Packet Attack Prevention

To prevent malformed packet attacks, a BSC6960 checks whether received packets have the characteristics of malformed packets.

The types of malformed packets that can be prevented include TCP local area network denial (LAND) and IP packets with incorrect field values. A BSC6960 directly discards such packets after identification. Currently, certain information about discarded packets can be queried by running the **DSP INVALIDPKTINFO** command with **DRPPKTTYPE** set to **MALFORM**. The information includes incorrect IP version number, incorrect IPv4 header length, all-0 IPv4 destination IP address, and broadcast IPv4 source IP address.

Malformed packet prevention starts automatically after BSC6960 startup and cannot be configured.

4.1.3.2.10 LACP Flood Attack Prevention

The following settings enable LACP flood attack prevention and alarm reporting on the BSC6960 side:

- **FLOODDEFEND.FLDTYPE** set to **LACP**
- **FLOODDEFEND.DFDSW** set to **ENABLE**
- **FLOODDEFEND.ALMSW** set to **ENABLE**

NOTE

It is recommended that LACP flood attack prevention and alarm reporting be enabled. If LACP flood attack prevention is disabled, the BSC6960 cannot defend against LACP flood attacks. If the LACP flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the BSC6960 encounters an LACP flood attack.

The BSC6960 checks the following items every 10 seconds:

- If the number of LACP flood packets received per second is greater than or equal to the value of **FLOODDEFEND.DFDTHD**, the packets are discarded and ALM-21665 Invalid Packets Exceeding Alarm is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of the **FLOODDEFEND.DFDTHD** parameter for five consecutive minutes, this alarm is cleared.

4.1.3.2.11 GTP-U Error Indication Flood Attack Defense

The BSC6960 classifies GTP-U packets to be sent for control-plane processing into two types: GTP-U Error Indication packets and GTP-U control packets, and provides different flood attack prevention thresholds.

The following settings enable GTP-U Error Indication flood attack prevention and alarm reporting on the BSC6960 side:

- **FLOODDEFEND.FLDTYPE** set to **GTPU_ERR_IND**
- **FLOODDEFEND.DFDSW** set to **ENABLE**
- **FLOODDEFEND.ALMSW** set to **ENABLE**

 NOTE

It is recommended that GTP-U Error Indication flood attack prevention and alarm reporting be enabled. If GTP-U Error Indication flood attack prevention is disabled, the BSC6960 cannot defend against GTP-U Error Indication flood attacks. If the GTP-U Error Indication flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the BSC6960 encounters a GTP-U Error Indication flood attack.

The BSC6960 checks the following items every 10 seconds:

- If the number of GTPU Error Indication flood packets received per second is greater than or equal to the value of **FLOODDEFEND.DFDTHD**, the packets are discarded and ALM-21665 Invalid Packets Exceeding Alarm is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of the **FLOODDEFEND.DFDTHD** parameter for five consecutive minutes, this alarm is cleared.

4.1.3.2.12 GTP-U Control Flood Attack Defense

The BSC6960 classifies GTP-U packets to be sent for control-plane processing into two types: GTP-U Error Indication packets and GTP-U control packets, and provides different flood attack prevention thresholds. The types of GTP-U control packets include echo requests, echo responses, notifications for supported extension headers, and 3GPP-reserved messages.

The following settings enable GTP-U control flood attack prevention and alarm reporting on the BSC6960 side:

- **FLOODDEFEND.FLDTYPE** set to **GTPU_CTRL**
- **FLOODDEFEND.DFDSW** set to **ENABLE**
- **FLOODDEFEND.ALMSW** set to **ENABLE**

 NOTE

It is recommended that GTP-U control flood attack prevention and alarm reporting be enabled. If GTP-U control flood attack prevention is disabled, the BSC6960 cannot defend against GTP-U control flood attacks. If the GTP-U control flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the BSC6960 encounters a GTP-U control flood attack.

The BSC6960 checks the following items every 10 seconds:

- If the number of GTPU flood packets received per second is greater than or equal to the value of **FLOODDEFEND.DFDTHD**, the packets are discarded and ALM-21665 Invalid Packets Exceeding Alarm is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of the **FLOODDEFEND.DFDTHD** parameter for five consecutive minutes, this alarm is cleared.

4.1.3.2.13 1588 L2 Multicast Packet Flood Attack Defense

The following settings enable 1588 L2 multicast packet flood attack defense and alarm reporting on the BSC6960 side:

- **FLOODDEFEND.FLDTYPE** set to **1588V2_L2_MULTICAST**
- **FLOODDEFEND.DFDSW** set to **ENABLE**
- **FLOODDEFEND.ALMSW** set to **ENABLE**

NOTE

It is recommended that 1588 L2 multicast packet flood attack defense and alarm reporting be enabled. If 1588 L2 multicast packet flood attack defense is disabled, the BSC6960 cannot defend against such attacks. If the 1588 L2 multicast packet flood alarm reporting function is disabled, the administrator cannot receive an alarm in a timely manner when the BSC6960 encounters such attacks.

The BSC6960 checks the following items every 10 seconds:

- If the number of 1588 L2 multicast flood packets received per second is greater than or equal to the value of **FLOODDEFEND.DFDTHD**, the packets are discarded and ALM-21665 Invalid Packets Exceeding Alarm is reported.
- After this alarm is generated, if the number of flood packets received per second is less than the value of the **FLOODDEFEND.DFDTHD** parameter for five consecutive minutes, this alarm is cleared.

4.2 Network Analysis

4.2.1 Benefits

The integrated firewall filters attack packets to improve equipment security.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.1.3 Function Impacts](#), [4.3.2.2.3 Function Impacts](#), [4.3.2.3.3 Function Impacts](#), [4.3.2.4.3 Function Impacts](#), [4.3.2.5.3 Function Impacts](#), [4.3.2.6.3 Function Impacts](#), [4.3.2.7.3 Function Impacts](#), [4.3.2.8.3 Function Impacts](#), [4.3.2.9.3 Function Impacts](#), and [4.3.2.10.3 Function Impacts](#).

Only the following functions have impacts on the network:

ARP/ND Spoofing Prevention

- If **IPGUARD.ARPLRNSTRICTPOLICY (LTE eNodeB, 5G gNodeB)** is set to **ONLY_RESPONSE** and an ARP entry corresponding to a packet already exists, the NE does not update the entry after receiving an ARP request packet but updates the ARP entry after receiving an ARP response packet. Entries may not be updated in a timely manner and services may be affected.
- If **IPGUARD.ARPLRNSTRICTPOLICY (LTE eNodeB, 5G gNodeB)** is set to **ALL**, the NE updates the ARP entries after receiving ARP request and response packets if the ARP entries corresponding to the packets already exist. In this case, the ARP entries of the NE are vulnerable to attacks and are frequently updated.

4.3 Requirements

4.3.1 Licenses

No license is required for the base station controller, eGBTS, NodeB, or gNodeB.

The operator must have purchased and activated the licenses for the features listed in the following table if the features are to be deployed for the eNodeB.

RA T	Feature ID	Feature Name	Model	License Control Item	NE	Sales Unit
LTE FD D	LOFD-0 03014	Integrated Firewall	LT1SINFIRE 00	Integrate d Firewall(FDD)	eNodeB	Per eNodeB
NB- IoT	MLOFD- 003014	Integrated Firewall	ML1SINFIR E00	Integrate d Firewall(NB-IoT)	eNodeB	Per eNodeB
LTE TD D	TDLOFD -003014	Integrated Firewall	LT1STINFIR 00	Integrate d Firewall(TDD)	eNodeB	Per eNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 LOFD-003014 Integrated Firewall

4.3.2.1.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Setting Notes
LTE FDD	Strict ARP learning	IPGUARD.ARPLRNSTRICTSW (LTE eNodeB, 5G gNodeB) set to ENABLE	<i>Equipment Security</i>	The IPGUARD.ARPLRNSTRICTPOLICY (LTE eNodeB, 5G gNodeB) parameter can be configured and take effect only when strict ARP learning is enabled.

4.3.2.1.2 Mutually Exclusive Functions

None

4.3.2.1.3 Function Impacts

None

4.3.2.2 LOFD-00301401 Access Control List (ACL)

4.3.2.2.1 Prerequisite Functions

None

4.3.2.2.2 Mutually Exclusive Functions

None

4.3.2.2.3 Function Impacts

None

4.3.2.3 LOFD-00301402 Automatic ACL Rule Configuration

4.3.2.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
LTE FDD	Access Control List (ACL)	None	<i>Equipment Security</i>

4.3.2.3.2 Mutually Exclusive Functions

None

4.3.2.3.3 Function Impacts

None

4.3.2.4 MLOFD-003014 Integrated Firewall

4.3.2.4.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Setting Notes
NB-IoT	Strict ARP learning	IPGUARD.ARPLRNSTRICTSW (LTE eNodeB, 5G gNodeB) set to ENABLE	<i>Equipment Security</i>	The IPGUARD.ARPLRNSTRICTPOLICY (LTE eNodeB, 5G gNodeB) parameter can be configured and take effect only when strict ARP learning is enabled.

4.3.2.4.2 Mutually Exclusive Functions

None

4.3.2.4.3 Function Impacts

None

4.3.2.5 MLOFD-00301401 Access Control List (ACL)

4.3.2.5.1 Prerequisite Functions

None

4.3.2.5.2 Mutually Exclusive Functions

None

4.3.2.5.3 Function Impacts

None

4.3.2.6 MLOFD-00301402 Automatic ACL Rule Configuration

4.3.2.6.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
NB-IoT	Access Control List (ACL)	None	<i>Equipment Security</i>

4.3.2.6.2 Mutually Exclusive Functions

None

4.3.2.6.3 Function Impacts

None

4.3.2.7 TDLOFD-003014 Integrated Firewall

4.3.2.7.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Setting Notes
LTE TDD	Strict ARP learning	IPGUARD.ARPLRNSTRICTSW (LTE eNodeB, 5G gNodeB) set to ENABLE	<i>Equipment Security</i>	The IPGUARD.ARPLRNSTRICTPOLICY (LTE eNodeB, 5G gNodeB) parameter can be configured and take effect only when strict ARP learning is enabled.

4.3.2.7.2 Mutually Exclusive Functions

None

4.3.2.7.3 Function Impacts

None

4.3.2.8 TDLOFD-00301401 Access Control List (ACL)

4.3.2.8.1 Prerequisite Functions

None

4.3.2.8.2 Mutually Exclusive Functions

None

4.3.2.8.3 Function Impacts

None

4.3.2.9 TDLOFD-00301402 Automatic ACL Rule Configuration

4.3.2.9.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
LTE TDD	Access Control List (ACL)	None	<i>Equipment Security</i>

4.3.2.9.2 Mutually Exclusive Functions

None

4.3.2.9.3 Function Impacts

None

4.3.2.10 FBFD-010023 Security Mechanism (Integrated Firewall)

4.3.2.10.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Setting Notes
NR FDD Low-frequency NR TDD High-frequency NR TDD	Strict ARP learning	IPGUARD.ARPLRNSTRICTSW (LTE eNodeB, 5G gNodeB) set to ENABLE	<i>Equipment Security</i>	The IPGUARD.ARPLRNSTRICTPOLICY (LTE eNodeB, 5G gNodeB) parameter can be configured and take effect only when strict ARP learning is enabled.

4.3.2.10.2 Mutually Exclusive Functions

None

4.3.2.10.3 Function Impacts

None

4.3.3 Hardware

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations (macro base stations)
UMTS	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

The base station must be configured with any of the following boards that provide Ethernet ports:

- Main control board: UMPT
- Extension transmission processing unit (UTRPc)

NOTE

A GBTS cannot be configured with a GTMU series board and a UTRPc.

- UMDU
- MDUC
- UBBPe/UBBPei/UBBPes/UBBPem/UBBPfw1/UBBPf3/UBBPg/UBBPph

Automatic IPv4 ACL rule configuration depends on the following:

- Main control board: UMPT
- UMPT+UTRPc
- UMDU
- MDUC

Automatic IPv6 ACL rule configuration depends on the following:

- UMPTg series boards
- UMPTga series boards
- UMPTHa series boards
- UMPTHb series boards

RF Modules

This function does not depend on RF modules.

4.3.4 Networking

- The base station controller must use IP over FE/GE/10GE transmission.
- The base station must use IP over FE/GE/10GE/25GE transmission.

4.3.5 Others

The current ACL rule specifications may be less than the sum of the specifications of the OM channel and clock, security, signaling, and service links. Therefore, obtain the number of configured OM channels and clock, security, signaling, and service links before automatic ACL rule configuration is enabled. Ensure that ACL rules to be automatically configured do not exceed the ACL rule specifications. If the number of configurable ACL rules is less than required, it is recommended that ACL rules be manually configured and the ACL rule matching scopes be set to network segments to reduce the number of ACL rules.

For a newly deployed or reconstructed separate-MPT co-transmission base station, the mode that provides the transmission port must be manually configured with ACL rules for data flows pertaining to the other mode to ensure service continuity, because automatic configuration of these ACL rules is not available. Similarly, for newly deployed or reconstructed cascaded base stations, ACL rules for data flows of a lower-level base station must be manually configured on an upper-level base station.

To activate automatic ACL rule configuration for an existing base station, note the following:

- If the base station has not been enabled with ACL-based packet filtering:
To ensure ongoing service continuity and signaling link connectivity, it is good practice to perform the following operations:
 - Before the activation, run the **ADD ACLRULE** command to configure an any-to-any ACL rule to allow all data flows to flow into the base station.
//Configuring an any-to-any ACL rule
ADD
ACLRULE:ACLID=3000,RULEID=1,PT=IP,SIP="0.0.0.0",SWC="255.255.255.255",DIP="0.0.0.0",DWC="255.255.255.255",MDSCP=NO;
 - After the activation, run the **RMV ACLRULE** command to delete the any-to-any ACL rule.
//Deleting an any-to-any ACL rule
RMV ACLRULE:ACLID=3000,RULEID=1;
- If the base station has been enabled with ACL-based packet filtering:
To ensure ongoing service continuity and signaling link connectivity, it is good practice to perform the following operations:
 - Before the activation, run the **ADD ACLRULE** command to configure an any-to-any ACL rule to allow all data flows to flow into the base station.
//Configuring an any-to-any ACL rule
ADD
ACLRULE:ACLID=3000,RULEID=1,PT=IP,SIP="0.0.0.0",SWC="255.255.255.255",DIP="0.0.0.0",DWC="255.255.255.255",MDSCP=NO;
 - Before activating automatic ACL rule configuration, delete ACL rules 1-59999 that can be automatically configured based on [4.1.1.2.3](#)

Automatically Configured ACL Rule Group. In addition, delete remaining ACL rules that are manually configured with the IDs ranging from 50000 to 59999 and reconfigure them with the IDs set to values ranging from 1 to 49999.

```
//Querying and recording ACL rules in the ID range of 1-59999, including automatically
configured ACL rules in the ID range of 1-59999 and manually configured ACL rules in the ID
range of 50000-59999
LST ACLRULE;;
//Deleting ACL rules in the ID range of 1-59999 that can be automatically configured and
remaining ACL rules that are manually configured in the ID range of 50000-59999
RMV ACLRULE:ACLID=3000,RULEID=50000;
//Reconfiguring the deleted ACL rules (in the ID range of 50000-59999) with the ID set to a
value ranging from 1 to 49999
ADD
ACLRULE:ACLID=3000,RULEID=1,PT=IP,SIP="1.1.1.1",SWC="255.255.255.255",DIP="2.2.2.2",DWC=
"255.255.255.255",MDSCP=NO;
```

- After the activation, run the **RMV ACLRULE** command to delete the any-to-any ACL rule.

```
//Deleting an any-to-any ACL rule
RMV ACLRULE:ACLID=3000,RULEID=1;
```

4.4 Operation and Maintenance (Base Station)

4.4.1 When to Use

It is good practice to enable ACL-based packet filtering and automatic ACL rule configuration for both signaling links and service links. For an existing base station that has been enabled with ACL-based packet filtering, it is recommended that automatic ACL rule configuration also be enabled.

4.4.2 Data Configuration

4.4.2.1 Data Preparation

"-" in the following tables indicates that there is no special requirement for setting the parameter. Set the parameter based on site requirements.

ACL-based Packet Filtering

[Table 4-5](#), [Table 4-6](#), [Table 4-7](#), [Table 4-8](#), [Table 4-9](#), and [Table 4-10](#) describe key parameters that must be set in the **ACL**, **ACLRULE**, **PACKETFILTER** (old model)/**PACKETFILTERING** (new model), **FTPCLTPORT**, and **EPGROUP** MOs, before activating ACL-based packet filtering.

Table 4-5 Data to be configured in the **ACL** MO

Parameter Name	Parameter ID	Setting Notes
ACL ID	ACL.ACLID (<i>5G gNodeB, LTE eNodeB</i>)	-

Parameter Name	Parameter ID	Setting Notes
Description	ACL.ACLDESC (5G gNodeB, LTE eNodeB)	-

Table 4-6 Data to be configured in the **ACLRULE** MO

Parameter Name	Parameter ID	Setting Notes
ACL ID	ACLRULE.ACLID (5G gNodeB, LTE eNodeB)	Set this parameter to the same value as that of the ACL.ACLID (LTE eNodeB, 5G gNodeB) parameter.
Rule ID	ACLRULE.RULEID (5G gNodeB, LTE eNodeB)	An ACL rule must have a unique ID.
Action	ACLRULE.ACTION (5G gNodeB, LTE eNodeB)	This parameter can be set to DENY or PERMIT . If an incoming packet matches an ACL rule, the base station determines whether to accept or discard the packet based on the value of this parameter. In the event of mismatch, the base station tries the next ACL rule until all rules in the ACL have been tried. Packets that do not match any ACL rule is processed based on the setting of the MB parameter in the MO associated with packet filtering.
Protocol Type	ACLRULE.PT (5G gNodeB, LTE eNodeB)	-
Source IP Address	ACLRULE.SIP (5G gNodeB, LTE eNodeB)	-
Destination IP Address	ACLRULE.DIP (5G gNodeB, LTE eNodeB)	-
Source Wildcard	ACLRULE.SWC (5G gNodeB, LTE eNodeB)	Set this parameter to 255.255.255.255 if an any-to-any ACL rule is required.
Destination Wildcard	ACLRULE.DWC (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID	Setting Notes
Match Source Port	ACLRULE.SMPT (5G gNodeB, LTE eNodeB)	This parameter specifies whether to check the source port number of each packet. This parameter is valid only if the PT parameter is set to TCP , UDP , or SCTP .
Source Port Operate	ACLRULE.SOP (5G gNodeB, LTE eNodeB)	-
Source Port 1	ACLRULE.SPT1 (5G gNodeB, LTE eNodeB)	This parameter is required only if the SMPT parameter is set to YES .
Source Port 2	ACLRULE.SPT2 (5G gNodeB, LTE eNodeB)	This parameter is required only if the SOP parameter is set to OP_RANGE .
Match Destination Port	ACLRULE.DMPT (5G gNodeB, LTE eNodeB)	This parameter specifies whether to check the destination port number of each packet. This parameter is valid only if the ACLRULE.PT (LTE eNodeB, 5G gNodeB) parameter is set to TCP , UDP , or SCTP .
Destination Port Operate	ACLRULE.DOP (5G gNodeB, LTE eNodeB)	-
Destination Port 1	ACLRULE.DPT1 (5G gNodeB, LTE eNodeB)	This parameter is required only if the DMPT parameter is set to YES .
Destination Port 2	ACLRULE.DPT2 (5G gNodeB, LTE eNodeB)	This parameter is required only if the DOP parameter is set to OP_RANGE .
Match DSCP	ACLRULE.MDSCP (5G gNodeB, LTE eNodeB)	-
DSCP	ACLRULE.DSCP (5G gNodeB, LTE eNodeB)	-

Parameter Name	Parameter ID	Setting Notes
VLAN ID Operate	ACLRULE.VLANIDOP (5G gNodeB, LTE eNodeB)	This parameter specifies the filtering criteria for VLAN IDs. This parameter is valid only if the value of the ACLRULE.ACLID (LTE eNodeB, 5G gNodeB) parameter ranges from 4000 to 4999. If this parameter is set to OP_NOVLAN, the ACLRULE.VLANID1 (LTE eNodeB, 5G gNodeB) and ACLRULE.VLANID2 (LTE eNodeB, 5G gNodeB) parameters cannot be set.
VLAN ID 1	ACLRULE.VLANID1 (5G gNodeB, LTE eNodeB)	• If the ACLRULE.VLANIDOP (LTE eNodeB, 5G gNodeB) parameter is set to OP_EQ, this parameter specifies the exact VLAN ID used for filtering. • If the ACLRULE.VLANIDOP (LTE eNodeB, 5G gNodeB) parameter is set to OP_RANGE, this parameter specifies the minimum VLAN ID used for filtering.
VLAN ID 2	ACLRULE.VLANID2 (5G gNodeB, LTE eNodeB)	If the ACLRULE.VLANIDOP (LTE eNodeB, 5G gNodeB) parameter is set to OP_RANGE, this parameter specifies the maximum VLAN ID used for filtering.

Table 4-7 lists the data to be prepared for the **PACKETFILTER** MO (old model) when the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**.

Table 4-7 Data to be configured in the **PACKETFILTER** MO (old model)

Parameter Name	Parameter ID	Setting Notes
Port Type	PACKETFILTER.PT (5G gNodeB, LTE eNodeB)	This parameter specifies the type of the port to which an ACL applies. The values of this parameter are as follows: • The value ETH indicates an Ethernet port, which cannot be a member of an Ethernet link aggregation group. In addition, port security must be enabled for this Ethernet port. • The value ETHTRK indicates an Ethernet link aggregation group. • The value ETHCI indicates a port for interconnection.
Port No.	PACKETFILTER.PN (5G gNodeB, LTE eNodeB)	-
Filter Mode	PACKETFILTER.FM (5G gNodeB, LTE eNodeB)	-
Advanced ACL ID	PACKETFILTER.ACLID (5G gNodeB, LTE eNodeB)	-
Layer 2 ACL ID	PACKETFILTER.ACLID2 (5G gNodeB, LTE eNodeB)	-
Match Behavior	PACKETFILTER.MB (5G gNodeB, LTE eNodeB)	-
ACL Rule Automatic Setup and Deletion Switch	PACKETFILTER.ACLAUTO SWITCH (5G gNodeB, LTE eNodeB)	If ACL-based packet filtering is enabled and simple service-related configurations are expected, it is recommended that this parameter be set to ON. In addition, the EPGROUP.PACKETFILTERSWITCH (LTE eNodeB, 5G gNodeB) parameter needs to be set to ENABLE.
OMCH Peer IP Limit Switch	PACKETFILTER.OMPEERIP LIMITSW (5G gNodeB, LTE eNodeB)	The value ON is recommended. If this switch is turned off, users can use any IP address to access the OM IP address of the base station, in which case, the base station may be attacked.

 NOTE

If a base station needs to allow packets that do not carry VLAN tags to pass through, the **PACKETFILTER.FM (LTE eNodeB, 5G gNodeB)** parameter is set to **L2_ACL** or **ADV_AND_L2**, and the **PACKETFILTER.MB (LTE eNodeB, 5G gNodeB)** parameter is set to **DENY**, it is required that the ACL specified by the **PACKETFILTER.ACLID2 (LTE eNodeB, 5G gNodeB)** parameter contains at least one ACL rule meeting both of the following conditions:

- The **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** parameter is set to **PERMIT** for the ACL rule.
- The **ACLRULE.VLANIDOP (5G gNodeB, LTE eNodeB)** parameter is set to **OP_NOVLAN** for the ACL rule.

Table 4-8 describes the data to be prepared when the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**.

Table 4-8 Data to be configured in the **PACKETFILTERING** MO (new model)

Parameter Name	Parameter ID	Setting Notes
Packet Filtering ID	PACKETFILTERING.PACKETFILTERINGID (5G gNodeB, LTE eNodeB)	-
Filter Mode	PACKETFILTERING.FM (5G gNodeB, LTE eNodeB)	-
Port Type	PACKETFILTERING.PT (5G gNodeB, LTE eNodeB)	This parameter specifies the type of the port to which an ACL applies. The values of this parameter are as follows: • The value ETH indicates an Ethernet port, which cannot be a member of an Ethernet link aggregation group. In addition, port security must be enabled for this Ethernet port. • The value ETHTRK indicates an Ethernet link aggregation group. • The value ETHCI indicates a port for interconnection.
Port ID	PACKETFILTERING.PORTID (5G gNodeB, LTE eNodeB)	-

Parameter Name	Parameter ID	Setting Notes
ACL ID	PACKETFILTERING.ACLID (5G gNodeB, LTE eNodeB)	-
Match Behavior	PACKETFILTERING.MB (5G gNodeB, LTE eNodeB)	-
ACL Rule Automatic Setup and Deletion Switch	PACKETFILTERING.ACLAUTOSWITCH (5G gNodeB, LTE eNodeB)	If ACL-based packet filtering is enabled and simple service-related configurations are expected, it is recommended that this parameter be set to ON . In addition, the EPGROUP.PACKETFILTERSWITCH (LTE eNodeB, 5G gNodeB) parameter needs to be set to ENABLE .
OMCH Peer IP Limit Switch	PACKETFILTERING.OMPEERLIMITSW (5G gNodeB, LTE eNodeB)	The value ON is recommended. If this switch is turned off, users can use any IP address to access the OM IP address of the base station, in which case, the base station may be attacked.

Table 4-9 Data to be configured in the **FTPCLTPORT** MO

Parameter Name	Parameter ID	Setting Notes
Start Number of the Data Port Range	FTPCLTPORT.STARTDATA PORT (5G gNodeB, LTE eNodeB)	Set this parameter as required. - When ACL Rule Automatic Setup and Deletion Switch is turned on, the automatically configured ACL rules contain the port range. - When ACL Rule Automatic Setup and Deletion Switch is turned off, the port range can be manually configured in ACL rules.
End Number of the Data Port Range	FTPCLTPORT.ENDDATA PORT (5G gNodeB, LTE eNodeB)	

Table 4-10 Data to be configured in the **EPGROUP** MO

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	EPGROUP.EPGROUPID (5G gNodeB, LTE eNodeB)	-
Packet Filter ACL Rule Auto-Setup-Deletion SW	EPGROUP.PACKETFILTER SWITCH (5G gNodeB, LTE eNodeB)	Set this parameter to ENABLE .

The source IP address and destination IP address in an ACL rule are used to distinguish data flows. [Table 4-11](#) describes the mapping between data flow types and source/destination IP addresses in ACL rules.

Table 4-11 Mapping between data types and source/destination IP addresses

Data Flow Types	Source IP Address	Destination IP Address
Control-plane or user-plane data flows	Peer IP address of the control plane or user plane	Local IP address of the control plane or user plane
Data flows (TCP, FTP, or SNTP packets) on the maintenance plane between a base station and the MAE	IP address of the MAE	Local OM IP address
ICMP data flows, for example, IP ping packets between a base station and a gateway	IP address of the gateway	

Data Flow Types	Source IP Address	Destination IP Address
Data flows on the maintenance plane between a base station and the NTP server	IP address of the NTP server	
IEEE 1588v2 clock packets from the active clock server	IP address of the active clock server	Local IP address of the IP clock
IEEE 1588v2 clock packets from the standby clock server	IP address of the standby clock server	
BFD data flows	Peer IP address of a BFD session	Local IP address of a BFD session

Table 4-12 describes key parameters that must be configured in the **IPGUARD** MO to enable alarm reporting for invalid packets exceeding a threshold in packet filtering.

Table 4-12 Data to be configured before activating alarm reporting for invalid packets exceeding a threshold

Parameter Name	Parameter ID	Setting Notes
Invalid Packet Exceeded Alarm Switch	IPGUARD.INVALIDPKTCH KSW (5G gNodeB, LTE eNodeB)	ENABLE is recommended. If this parameter is set to DISABLE , invalid packet attack defense will be disabled, and the base station may encounter invalid packet attacks.
Invalid Packet Alarm Threshold	IPGUARD.INVALIDPKTAL MTHD (5G gNodeB, LTE eNodeB)	The default values are recommended.

Flood Attack Defense

Table 4-13 describes key parameters that must be set in the **FLOODDEFEND** MO.

Table 4-13 Data to be configured before activating flood attack defense

Parameter Name	Parameter ID	Setting Notes
Flood Type	FLOODDEFEND.FLDTYPE (5G gNodeB, LTE eNodeB)	Set this parameter based on the flood attack analysis results.
Defend Switch	FLOODDEFEND.DFDSW (5G gNodeB, LTE eNodeB)	Set this parameter to ENABLE . If this parameter is set to DISABLE , flood attack defense of the corresponding type will be disabled, and the base station may encounter flood attacks.
Defend Threshold	FLOODDEFEND.DFDTHD (5G gNodeB, LTE eNodeB)	Set this parameter to its recommended value.
Alarm Switch	FLOODDEFEND.ALMSW (5G gNodeB, LTE eNodeB)	Set this parameter to ENABLE if alarms need to be reported when flood packets are launched. If this parameter is set to DISABLE , the corresponding flood attack alarm reporting function is disabled, and the administrator cannot obtain attack warnings in a timely manner.

ARP Spoofing Prevention

Table 4-14 describes key parameters that must be configured in the **IPGUARD** MO to enable ARP spoofing prevention.

Table 4-14 Data to be configured before activating ARP spoofing prevention

Parameter Name	Parameter ID	Setting Notes
ARP Spoofing Check Switch	IPGUARD.ARPSPOOFCHK SW (5G gNodeB, LTE eNodeB)	Set this parameter to ENABLE . If this parameter is set to DISABLE , ARP spoofing prevention will be disabled, and the base station may encounter ARP spoofing attacks.
ARP Spoofing Alarm Threshold	IPGUARD.ARPSPOOFALM THD (5G gNodeB, LTE eNodeB)	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
ARP Learning Strict Switch	IPGUARD.ARPLRNSTRICT SW (5G gNodeB, LTE eNodeB)	Set this parameter to ENABLE to enable strict ARP learning. If this parameter is set to DISABLE, strict ARP learning will be disabled. In this case, the ARP entries of the base station may be exhausted by attackers.
ARP Learning Strict Policy	IPGUARD.ARPLRNSTRICT POLICY (LTE eNodeB, 5G gNodeB)	This parameter can be configured and take effect only when strict ARP learning is enabled. Set this parameter to ONLY_RESPONSE when ARP entries need to be updated only after ARP response packets are received. • If this parameter is set to ONLY_RESPONSE and an ARP entry corresponding to a packet already exists, the base station does not update the entry after receiving an ARP request packet but updates the ARP entry after receiving an ARP response packet. Entries may not be updated in a timely manner, affecting services. • If this parameter is set to ALL, the base station updates the ARP entries after receiving ARP request and response packets if the ARP entries corresponding to the packets already exist. In this case, the ARP entries of the base station are vulnerable to attacks and are frequently updated.

Invalid Packet Attack Prevention

Invalid packet attack prevention is enabled automatically.

ICMP Response Attack Prevention

Table 4-15 describes key parameters that must be configured in the **TRANSFUNCTIONSW** MO to enable ICMP response attack prevention.

Table 4-15 Data to be configured before activating ICMP response attack prevention

Parameter Name	Parameter ID	Setting Notes
ICMP Port Unreachable Send Switch	TRANSFUNCTIONSW.ICMPPORTUNREACHABLESW (5G gNodeB, LTE eNodeB)	DISABLE is recommended. If this parameter is set to ENABLE , the base station will allow sending of ICMP destination unreachable response messages for a long period of time. As a result, attackers may obtain the network topology of the base station.

4.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

- Activating ACL-based packet filtering (manual ACL rule configuration in IPv4)
//Configuring an ACL
ADD ACL:ACLID=3000,ACLDISC="Acl Group is created";
//Configuring an ACL rule that uses IP as the protocol type and allows communication with the 192.168.5.0 network segment
ADD
ACLRULE:ACLID=3000,RULEID=10,ACTION=PERMIT,PT=IP,SIP="192.168.5.5",SWC="0.0.0.255",DIP="192.168.5.6",DWC="0.0.0.255",MDSCP=YES,DSCP=25;
//Configuring an ACL rule that uses ICMP as the protocol type and renders 192.168.5.5 pingable.
Keeping this ACL rule poses security risks. It is good practice to delete this rule after the ping operation is complete.
ADD
ACLRULE:ACLID=3000,RULEID=11,ACTION=PERMIT,PT=ICMP,SIP="192.168.5.5",SWC="0.0.0.0",DIP="192.168.5.6",DWC="0.0.0.0",MDSCP=YES,DSCP=25;
//Configuring ACL-based packet filtering without automatic ACL rule configuration when **GTRANSPARA.TRANSFCGMODE** is set to **OLD**
ADD
PACKETFILTER:CN=0,SRN=0,SN=6,SBT=BASE_BOARD,PT=ETH,PN=0,FM=ADV_ACL,ACLID=3000,MB=DENY,ACLAUTOSWITCH=OFF;
//Configuring ACL-based packet filtering without automatic ACL rule configuration when **GTRANSPARA.TRANSFCGMODE** is set to **NEW**
ADD PACKETFILTERING: PACKETFILTERINGID=0, FM=ADV_ACL, ACLTYPE=IPV4, PT=INTERFACE, PORTID=60, ACLID=3000, MB=DENY, ACLAUTOSWITCH=OFF;
//Activating alarm reporting for invalid packets exceeding a threshold
SET IPGUARD:INVALIDPKTCHKSW=ENABLE,INVALIDPKTALMTHD=2000;
- Activating ACL-based packet filtering (manual ACL rule configuration in IPv6)
//Configuring an ACL
ADD ACL6:ACLID=3001,ACLDISC="Acl Group is created";

```
//Configuring an ACL rule that uses IP as the protocol type and allows communication with the
2002::10:0:0 network segment
ADD
ACLRULE6:ACLID=3001,RULEID=10,ACTION=PERMIT,PT=IP6,SIP="2002::10:0:1",SPFXLEN=96,DIP="0::0",
DPFXLEN=0,MDSCP=YES,DSCP=25;
//Configuring an ACL rule (IPv6 is based on the ND sub-protocol of ICMP6. Therefore, you must
enable the ICMP6 protocol between the interface address and the next-hop address so that the base
station can receive ND packets.)
ADD
ACLRULE6:ACLID=3001,RULEID=11,ACTION=PERMIT,PT=ICMP6,SIP="2002::11:0:1",SPFXLEN=96,DIP="2
002::11:0:2",DPFXLEN=96,MDSCP=YES,DSCP=25;
//Configuring ACL-based packet filtering without automatic ACL rule configuration
ADD PACKETFILTERING:PACKETFILTERINGID=0, FM=ADV_ACL, ACLTYPE=IPV6, PT=INTERFACE,
PORTID=60, ACLID=3001, MB=DENY, ACLAUTOSWITCH=OFF;
//Activating alarm reporting for invalid packets exceeding a threshold
SET IPGUARD:INVALIDPKTCHKSW=ENABLE,INVALIDPKTALMTHD=2000;
```

- Activating ACL-based packet filtering (automatic ACL rule configuration)
 - For a newly deployed base station


```
//Configuring an ACL
ADD ACL:ACLID=3000,ACLDISC="Acl Group is created";
//Configuring ACL-based packet filtering with automatic ACL rule configuration when
GTRANSPARA.TRANSFCFGMODE is set to OLD
ADD
PACKETFILTER:CN=0,SRN=0,SN=6,SBT=BASE_BOARD,PT=ETH,PN=0,FM=ADV_ACL,ACLID=3000,M
B=DENY,ACLAUTOSWITCH=ON;
//Configuring ACL-based packet filtering with automatic ACL rule configuration when
GTRANSPARA.TRANSFCFGMODE is set to NEW
ADD PACKETFILTERING:PACKETFILTERINGID=0,FM=ADV_ACL, ACLTYPE=IPV4, PT=INTERFACE,
PORTID=60, ACLID=3000,MB=DENY,ACLAUTOSWITCH=ON;
//Configuring automatic ACL rule configuration in endpoint mode
ADD EPGROUP:EPGROUPEID=1,PACKETFILTERSWITCH=ENABLE;
```
 - For an existing base station that has been enabled with ACL-based packet filtering


```
//Changing ACL-based packet filtering configurations to enable automatic ACL rule
configuration when GTRANSPARA.TRANSFCFGMODE is set to OLD
MOD PACKETFILTER:CN=0,SRN=0,SN=6,SBT=BASE_BOARD,PT=ETH,PN=0,ACLAUTOSWITCH=ON;
//Changing ACL-based packet filtering configurations to enable automatic ACL rule
configuration when GTRANSPARA.TRANSFCFGMODE is set to NEW
ADD PACKETFILTERING:PACKETFILTERINGID=0,FM=ADV_ACL, ACLTYPE=IPV4, PT=INTERFACE,
PORTID=60, ACLID=3000,ACLAUTOSWITCH=ON;
//Configuring automatic ACL rule configuration in endpoint mode
MOD EPGROUP:EPGROUPEID=1,PACKETFILTERSWITCH=ENABLE;
```
 - For an existing base station that has not been enabled with ACL-based packet filtering


```
//Configuring an ACL
ADD ACL:ACLID=3000,ACLDISC="Acl Group is created";
//Configuring ACL-based packet filtering with automatic ACL rule configuration when
GTRANSPARA.TRANSFCFGMODE is set to OLD
ADD
PACKETFILTER:CN=0,SRN=0,SN=6,SBT=BASE_BOARD,PT=ETH,PN=0,FM=ADV_ACL,ACLID=3000,M
B=DENY,ACLAUTOSWITCH=ON;
//Configuring ACL-based packet filtering with automatic ACL rule configuration when
GTRANSPARA.TRANSFCFGMODE is set to NEW
ADD PACKETFILTERING:PACKETFILTERINGID=0,FM=ADV_ACL, ACLTYPE=IPV4, PT=INTERFACE,
PORTID=60, ACLID=3000,MB=DENY,ACLAUTOSWITCH=ON;
//Configuring automatic ACL rule configuration in endpoint mode
MOD EPGROUP:EPGROUPEID=1,PACKETFILTERSWITCH=ENABLE;
```
- Activating other security functions


```
//Activating flood attack prevention
ADD FLOODDEFEND:FLDTYPE=ARP,DFDSW=ENABLE,DFDTHD=512,ALMSW=ENABLE;
//Activating ARP spoofing prevention
SET IPGUARD:ARPSPOOFCHKS=ENABLE,ARPSPOOFALMTHD=120,ARPLRNSTRICTSW=ENABLE;
//(Optional) Configuring strict ARP learning so that ARP entries are updated only after receiving ARP
reply packets.
SET IPGUARD:ARPLRNSTRICTSW=ENABLE, ARPLRNSTRICTPOLICY=ONLY_RESPONSE;
//Activating ND spoofing prevention
```

```
SET IPGUARD:NDSPPOOFCHKSW=ENABLE,NDSPPOOFALMTHD=120;  
//Activating ICMP response attack prevention  
SET TRANSFUNCTIONSW: ICMPPORTUNREACHABLESW=DISABLE;
```

Deactivation Command Examples

- Deactivating automatic ACL rule configuration and retaining ACL-based packet filtering for a port

```
ADD  
ACLRULE:ACLID=3001,RULEID=1,PT=IP,SIP="0.0.0.0",SWC="0.0.0.0",DIP="0.0.0.0",DWC="0.0.0.0",MDSCP  
=NO;  
MOD EPGROUP:EPGROUPID=100,PACKETFILTERSWITCH=DISABLE;  
//When GTRANSPARA.TRANSFCFGMODE is set to OLD  
MOD PACKETFILTER:SN=7,SBT=BASE_BOARD,PT=ETH,PN=1,ACLAUTOSWITCH=OFF;  
//When GTRANSPARA.TRANSFCFGMODE is set to NEW  
MOD PACKETFILTERING:PACKETFILTERINGID=2,FM=ADV_ACL, ACLTYPE=IPV4, PT=INTERFACE,  
PORTID=60, ACLID=3000,ACLAUTOSWITCH=OFF;  
ADD  
ACLRULE:ACLID=3001,RULEID=2,PT=IP,SIP="1.1.1.1",SWC="0.0.0.0",DIP="2.2.2.2",DWC="0.0.0.0",MDSCP  
=NO;  
RMV ACLRULE:ACLID=3001,RULEID=1;
```

- Deactivating both automatic ACL rule configuration and ACL-based packet filtering for a port

```
ADD  
ACLRULE:ACLID=3001,RULEID=1,PT=IP,SIP="0.0.0.0",SWC="0.0.0.0",DIP="0.0.0.0",DWC="0.0.0.0",MDSCP  
=NO;  
MOD EPGROUP:EPGROUPID=100,PACKETFILTERSWITCH=DISABLE;  
//When GTRANSPARA.TRANSFCFGMODE is set to OLD  
RMV PACKETFILTER: CN=0,SRN=0,SN=6,SBT=BASE_BOARD,PT=ETH,PN=0;  
//When GTRANSPARA.TRANSFCFGMODE is set to NEW  
RMV PACKETFILTERING:PACKETFILTERINGID=2;  
RMV ACLRULE:ACLID=3001,RULEID=1;  
RMV ACL:ACLID=3001;
```

- Deactivating other security functions

```
//Deactivating flood attack prevention. Deactivating flood attack prevention requires removing  
configurations for all types of attack packets. The example in this section is specific to ARP packets  
only.  
RMV FLOODDEFEND:FLDTYPE=ARP;  
//Deactivating alarm reporting for invalid packets exceeding a threshold  
SET IPGUARD:INVALIDPKTCHKSW=DISABLE;  
//Deactivating ARP spoofing prevention  
SET IPGUARD:ARPSPOOFCHKSW=DISABLE, ARPLRNSTRICTSW=DISABLE;  
//Deactivating ND spoofing prevention  
SET IPGUARD:NDSPPOOFCHKSW=DISABLE;  
//Deactivating ICMP response attack prevention  
SET TRANSFUNCTIONSW: ICMPPORTUNREACHABLESW=ENABLE;
```

4.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

The activation observation procedures in this section are applicable when network attacks exist.

ACL-based Packet Filtering

Step 1 Verify that permitted packets are properly received.

Use SCTP packets over the S1 interface as an example. Run the **DSP SCTPLNK** command. If the value of **SCTP Link Status** is **Up** in the command output, the function is activated.

Step 2 Verify that denied packets are discarded.

Use a device whose packets should be denied to ping the interface IP address of the base station. If the device receives a response indicating a ping failure, ACL-based packet filtering has been activated.

----End

Automatic ACL Rule Configuration

- If the **OMCH**, **SCTPLNK**, **IPPATH**, **IPCLKLNK**, **NTPC**, **CA**, **IKEPEER**, **CRLTSK**, and **EPGRP** MOs have been configured, run the **DSP ACLRULE** command to check for automatically configured ACL rules for signaling packets, service packets, OM packets, IPsec packets in secure scenarios, and clock packets.
- To verify this function for the FTP user plane, perform base station file download or upload on the MAE and run the **DSP ACLRULE** command to check for automatically configured ACL rules.

Flood Attack Prevention

Run the **DSP FLOODDEFEND** command. If the value of **Number of Flood Dropped Packets** is not **0** in the command output, flood attack prevention is activated.

Invalid Packet Alarm Reporting

Run the **DSP INVALIDPKTINFO** command. If invalid packet information is available in the command output, invalid packet alarm reporting is successfully activated.

ARP Spoofing Prevention

Run the **DSP ARPSPOOFING** command. If IP address and MAC address information of ARP spoofed packets is available in the command output, ARP spoofing prevention is activated.

ICMP Response Attack Prevention

Check whether the peer device receives ICMP destination unreachable response packets. If no ICMP unreachable response packet is received, ICMP response attack prevention is activated.

4.4.4 Network Monitoring

None

4.5 Operation and Maintenance (BSC6900/BSC6910)

4.5.1 When to Use

It is recommended that ACL-based packet filtering be enabled in various scenarios.

4.5.2 Data Configuration

4.5.2.1 Data Preparation

"-" in the following tables indicates that there is no special requirement for setting the parameter. Set the parameter based on site requirements.

ACL-based Packet Filtering

Key parameters in the **ACL**, **ACLRULE**, and **PACKETFILTER** MOs must be configured to enable ACL-based packet filtering, as described in [Table 4-16](#), [Table 4-17](#), and [Table 4-18](#), respectively.

Table 4-16 Data to be configured in the **ACL** MO

Parameter Name	Parameter ID	Setting Notes
ACL ID	ACL.ACLID	-
Description	ACL.ACLDESC	-

Table 4-17 Data to be configured in the **ACLRULE** MO

Parameter Name	Parameter ID	Setting Notes
ACL ID	ACLRULE.ACLID	Set this parameter to the same value as ACL.ACLID .
Rule ID	ACLRULE.RULEID	-
Destination IP Address	ACLRULE.DIP	-

Table 4-18 Data to be configured in the **PACKETFILTER** MO

Parameter Name	Parameter ID	Setting Notes
Subrack No.	PACKETFILTER.SRN	-

Parameter Name	Parameter ID	Setting Notes
Slot No.	PACKETFILTER.SN	-
Port Type	PACKETFILTER.PORTTYPE	The values of this parameter are as follows: - The value ETHER indicates an Ethernet port, which cannot be a member of an Ethernet link aggregation group. - The value TRUNK indicates an Ethernet link aggregation group.
Port No.	PACKETFILTER.PN	This parameter specifies the number of a port.
ACL ID	PACKETFILTER.ACLID	Set this parameter to the same value as ACL.ACLID .

Rate Limitation on Broadcast Packets

Table 4-19 Data to be configured in the **ETHPORT MO**

Parameter Name	Parameter ID	Setting Notes
Received broadcast packets alarm threshold[pac ket/s]	ETHPORT.BCPKTALARMTHD	The default values are recommended.
Received broadcast packets alarm clear threshold[pac ket/s]	ETHPORT.BCPKTALARMC LRTHD	

ICMP Flood Attack Prevention

Key parameters in the **IPGUARD** and **ICMPGUARD** MOs must be configured to enable ICMP flood attack prevention, as described in [Table 4-20](#) and [Table 4-21](#), respectively.

Table 4-20 Data to be configured in the **IPGUARD** MO

Parameter Name	Parameter ID	Setting Notes
ICMP Attack Check Switch	IPGUARD.ICMPALMSW	Set this parameter to ON .
ICMP Attack Alarm Threshold	IPGUARD.ICMPALMTHD	The default values are recommended.
ICMP Attack Alarm Clearance Threshold	IPGUARD.ICMPALMRTHD	

Table 4-21 Data to be configured in the **ICMPGUARD** MO

Parameter Name	Parameter ID	Setting Notes
Packet source IP address	ICMPGUARD.IPADDR	-
ICMP Type to Guard Against	ICMPGUARD.GUARDTYPE	Set this parameter as required.

ARP Flood Attack Prevention

Table 4-22 Data to be configured in the **IPGUARD** MO

Parameter Name	Parameter ID	Setting Notes
ARP Learning Strict Switch	IPGUARD.ARPLRNSTRICTSW	Set this parameter to ON .

ARP Spoofing Prevention

Table 4-23 Data to be configured in the **IPGUARD** MO

Parameter Name	Parameter ID	Setting Notes
Gratuitous Arp Anti Cheat Switch	IPGUARD.ARPANTICHEATSW	Set this parameter to ON .

Illegal Packet Attack Prevention

Table 4-24 Data to be configured in the **IPGUARD** MO

Parameter Name	Parameter ID	Setting Notes
Invalid IP Packet Check Switch	IPGUARD.VALIDPKTCHKSW	Set these parameters to ON .
Invalid Multicast MAC Packet Check Switch	IPGUARD.INVALIDMCASTALMSW	
Invalid IP Packet Alarm Threshold	IPGUARD.INVALIDPKTALMTHD	The default values are recommended.
Invalid IP Packet Alarm Clearance Threshold	IPGUARD.INVALIDPKTALMRTHD	
Invalid Multicast MAC Packet Alarm Threshold	IPGUARD.INVALIDMCASTALMTHD	
Invalid Multicast MAC Packet Alarm Clearance Threshold	IPGUARD.INVALIDMCASTALMRTHD	

4.5.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Activating ACL-based packet filtering (manual ACL rule configuration)
//Configuring an ACL
```

```
ADD ACL: ACLID=100, ACLDESC="ACL Description";
//Configuring an ACL rule
ADD ACLRULE: ACLID=100, RULEID=1, DIP="192.168.5.6";
//Configuring ACL-based packet filtering
ADD PACKETFILTER: SRN=3, SN=8, PORTTYPE=ETHER, PN=0, ACLID=100;
//Activating other security functions
//Setting the alarm thresholds for rate limitation on broadcast packets
SET ETHPORT: SRN=3, SN=10, BRDTYPE=FG2c, PTYPE=FE, PN=0, BCPKTALARMTHD=1000,
BCPKTALARMCLRTHD=200;
//Enabling ICMP flood attack prevention
SET IPGUARD: SRN=3, SN=10, BRDTYPE=FG2c, ICMPALMSW=ON;
//Adding an ICMP flood attack prevention policy
ADD ICMPGUARD: SRN=3, SN=10, IPADDR="10.10.20.0", MASK="255.255.255.0",
GUARDTYPE=ECHOREPLY-0&UNREACH-0&SOURCEQUENCH-0&REDIRECT-0&ECHO-1&ROUTERADVERT-0&R
OUTERSOLICIT-0&TIMXCEED-0&PARAMPROB-0&TSTAMP-0&TSTAMPREPLY-0&IREQ-0&IREQREPLY-0&MASK
REQ-0&MASKREPLY-0;
//Enabling ARP flood attack prevention
SET IPGUARD: SRN=3, SN=10, BRDTYPE=FG2c, ARPLRNSTRICTSW=ON;
//Enabling ARP spoofing prevention
SET IPGUARD: SRN=3, SN=10, BRDTYPE=FG2c, ARPANTICHEATSW=ON;
//Activating illegal packet attack prevention
SET IPGUARD: SRN=3, SN=10, BRDTYPE=FG2c, VALIDPKTCHKSW=ON;
```

Deactivation Command Examples

```
//Deactivating ACL-based packet filtering for a port
RMV PACKETFILTER: SRN=3, SN=8, PORTTYPE=ETHER, PN=0;
//Deactivating ICMP flood attack prevention
SET IPGUARD: SRN=3, SN=10, BRDTYPE=FG2c, ICMPALMSW=OFF;
//Deactivating ARP flood attack prevention
SET IPGUARD: SRN=3, SN=10, BRDTYPE=FG2c, ARPLRNSTRICTSW=OFF;
//Deactivating ARP spoofing prevention
SET IPGUARD: SRN=3, SN=10, BRDTYPE=FG2c, ARPANTICHEATSW=OFF;
//Deactivating illegal packet attack prevention
SET IPGUARD: SRN=3, SN=10, BRDTYPE=FG2c, VALIDPKTCHKSW=OFF;
```

4.5.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.5.3 Activation Verification

The activation observation procedures in this section are applicable when network attacks exist.

ACL-based Packet Filtering

To verify ACL-based packet filtering, perform the following steps:

- Step 1** Verify that permitted packets are properly received. Use SCTP packets as an example. Run the **DSP SCTPLNK** command. If the value of **SCTP Link Status** is **Up** in the command output, the function is activated.
- Step 2** Verify that denied packets are discarded. Run the **DSP PACKETFILTER** command. If **Number of TX Deny Packets(packet)** is not **0** in the command output, the function is activated.

----End

Rate Limitation on Broadcast Packets

This function is enabled by default. Therefore, there is no need to verify it.

ICMP/ARP Flood Attack Prevention

Run the **LST IPGUARD** command to query whether the related policies are correctly configured.

ARP Spoofing Prevention

Run the **DSP ARPSPOOFING** command. If IP address and MAC address information of ARP spoofed packets is available in the command output, ARP spoofing prevention is activated.

Illegal Packet Alarm Reporting

Run the **DSP INVALIDPKTINFO** command. If illegal packet information is available in the command output, illegal packet alarm reporting is successfully activated.

4.5.4 Network Monitoring

None

4.6 Operation and Maintenance (BSC6960)

4.6.1 When to Use

It is recommended that ACL-based packet filtering be enabled in various scenarios.

4.6.2 Data Configuration

4.6.2.1 Data Preparation

"-" in the following tables indicates that there is no special requirement for setting the parameter. Set the parameter based on site requirements.

ACL-based Packet Filtering

[Table 4-25](#), [Table 4-26](#), and [Table 4-27](#) describe key parameters that must be set in the **ACL**, **ACLRULE**, and **PACKETFILTERING** MOs to enable ACL-based packet filtering.

Table 4-25 Data to be configured in the **ACL** MO

Parameter Name	Parameter ID	Setting Notes
ACL ID	ACL.ACLID	-

Parameter Name	Parameter ID	Setting Notes
Description	ACL.ACLDESC	-

Table 4-26 Data to be configured in the **ACLRULE** MO

Parameter Name	Parameter ID	Setting Notes
ACL ID	ACLRULE.ACLID	Set this parameter to the same value as that of the ACL.ACLID parameter.
Rule ID	ACLRULE.RULEID	An ACL rule must have a unique ID.
Action	ACLRULE.ACTION	This parameter can be set to DENY or PERMIT . If an incoming packet matches an ACL rule, the base station determines whether to accept or discard the packet based on the value of this parameter. In the event of mismatch, the base station tries the next ACL rule until all rules in the ACL have been tried. Packets that do not match any ACL rule is processed based on the setting of the MB parameter in the MO associated with packet filtering.
Protocol Type	ACLRULE.PT	-
Source IP Address	ACLRULE.SIP	-
Destination IP Address	ACLRULE.DIP	-
Source Wildcard	ACLRULE.SWC	Set this parameter to 255.255.255.255 if an any-to-any ACL rule is required.
Destination Wildcard	ACLRULE.DWC	
Match Source Port	ACLRULE.SMPT	This parameter specifies whether to check the source port number of each packet. This parameter is valid only if the PT parameter is set to TCP , UDP , or SCTP .

Parameter Name	Parameter ID	Setting Notes
Source Port Operate	ACLRULE.SOP	-
Source Port 1	ACLRULE.SPT1	This parameter is required only if the SMPT parameter is set to YES .
Source Port 2	ACLRULE.SPT2	This parameter is required only if the SOP parameter is set to OP_RANGE .
Match Destination Port	ACLRULE.DMPT	This parameter specifies whether to check the destination port number of each packet. This parameter is valid only if the ACLRULE.PT parameter is set to TCP , UDP , or SCTP .
Destination Port Operate	ACLRULE.DOP	-
Destination Port 1	ACLRULE.DPT1	This parameter is required only if the DMPT parameter is set to YES .
Destination Port 2	ACLRULE.DPT2	This parameter is required only if the DOP parameter is set to OP_RANGE .
Match DSCP	ACLRULE.MDSCP	-
DSCP	ACLRULE.DSCP	-
VLAN ID Operate	ACLRULE.VLANIDOP	This parameter specifies the filtering criteria for VLAN IDs. This parameter is valid only if the value of the ACLRULE.ACLID parameter ranges from 4000 to 4999. If this parameter is set to OP_NOVLAN , the ACLRULE.VLANID1 and ACLRULE.VLANID2 parameters cannot be set.
VLAN ID 1	ACLRULE.VLANID1	- If the ACLRULE.VLANIDOP parameter is set to OP_EQ, this parameter specifies the exact VLAN ID used for filtering. - If the ACLRULE.VLANIDOP parameter is set to OP_RANGE, this parameter specifies the minimum VLAN ID used for filtering.

Parameter Name	Parameter ID	Setting Notes
VLAN ID 2	ACLRULE.VLANID2	If the ACLRULE.VLANIDOP parameter is set to OP_RANGE , this parameter specifies the maximum VLAN ID used for filtering.

Table 4-27 Data to be configured in the **PACKETFILTERING** MO

Parameter Name	Parameter ID	Setting Notes
Packet Filtering ID	PACKETFILTERING.PACKETFILTERINGID	-
Filter Mode	PACKETFILTERING.FM	-
Port Type	PACKETFILTERING.PT	This parameter specifies the type of the port to which an ACL applies. The values of this parameter are as follows: • The value ETH indicates an Ethernet port, which cannot be a member of an Ethernet link aggregation group. In addition, port security must be enabled for this Ethernet port. • The value ETHTRK indicates an Ethernet link aggregation group. • The value INTERFACE indicates an interface.
Port ID	PACKETFILTERING.PORTID	-
ACL ID	PACKETFILTERING.ACLID	-
Match Behavior	PACKETFILTERING.MB	-

The source IP address and destination IP address in an ACL rule are used to distinguish data flows. [Table 4-28](#) describes the mapping between data flow types and source/destination IP addresses in ACL rules.

Table 4-28 Mapping between data types and source/destination IP addresses

Data Flow Types	Source IP Address	Destination IP Address
Control-plane or user-plane data flows	Peer IP address of the control plane or user plane	Local IP address of the control plane or user plane
Data flows (TCP, FTP, or SNMP packets) on the maintenance plane between a BSC6960 and the MAE	IP address of the MAE	Local OM IP address
ICMP data flows, for example, IP ping packets between a BSC6960 and a gateway	IP address of the gateway	
Data flows on the maintenance plane between a BSC6960 and the NTP server	IP address of the NTP server	
IEEE 1588V2 clock packets from the active clock server	IP address of the active clock server	Local IP address of the IP clock
IEEE 1588V2 clock packets from the standby clock server	IP address of the standby clock server	
BFD data flows	Peer IP address of a BFD session	Local IP address of a BFD session

Table 4-29 describes key parameters that must be configured in the **IPGUARD** MO to enable alarm reporting for invalid packets exceeding a threshold in packet filtering.

Table 4-29 Data to be configured before activating alarm reporting for invalid packets exceeding a threshold

Parameter Name	Parameter ID	Setting Notes
Invalid Packet Exceeded Alarm Switch	IPGUARD.INVALIDPKTCHKSW	ENABLE is recommended. If this parameter is set to DISABLE , invalid packet attack defense will be disabled, and the BSC6960 may encounter invalid packet attacks.
Invalid Packet Alarm Threshold	IPGUARD.INVALIDPKTALMTHD	The default value is recommended.

Flood Attack Defense

Table 4-30 describes key parameters that must be set in the **FLOODDEFEND** MO.

Table 4-30 Data to be configured before activating flood attack defense

Parameter Name	Parameter ID	Setting Notes
Flood Type	FLOODDEFEND.FLDTYPE	Set this parameter based on the flood attack analysis results.
Defend Switch	FLOODDEFEND.DFDSW	Set this parameter to ENABLE . If this parameter is set to DISABLE , flood attack defense of the corresponding type will be disabled, and the BSC6960 may encounter flood attacks.
Defend Threshold	FLOODDEFEND.DFDTHD	Set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Alarm Switch	FLOODDEFEND.ALMSW	Set this parameter to ENABLE if alarms need to be reported when flood packets are launched. If this parameter is set to DISABLE , the corresponding flood attack alarm reporting function is disabled, and the administrator cannot obtain attack warnings in a timely manner.

ARP Spoofing Prevention

Table 4-31 describes key parameters that must be configured in the **IPGUARD** MO to enable ARP spoofing prevention.

Table 4-31 Data to be configured before activating ARP spoofing prevention

Parameter Name	Parameter ID	Setting Notes
ARP Spoofing Check Switch	IPGUARD.ARPSPOOFCHK SW	Set this parameter to ENABLE . If this parameter is set to DISABLE , ARP spoofing prevention will be disabled, and the BSC6960 may encounter ARP spoofing attacks.
ARP Spoofing Alarm Threshold	IPGUARD.ARPSPOOFALM THD	The default value is recommended.
ARP Learning Strict Switch	IPGUARD.ARPLRNSTRICT SW	Set this parameter to ENABLE . If this parameter is set to DISABLE , strict ARP learning will be disabled. In this case, the ARP entries of the BSC6960 may be exhausted by attackers.

Invalid Packet Attack Prevention

Invalid packet attack prevention is enabled automatically.

ICMP Response Attack Prevention

Table 4-32 describes key parameters that must be configured in the **TRANSFUNCTIONSW** MO to enable ICMP response attack prevention.

Table 4-32 Data to be configured before activating ICMP response attack prevention

Parameter Name	Parameter ID	Setting Notes
ICMP Port Unreachable Send Switch	TRANSFUNCTIONSW.ICMPPORTUNREACHABLESW	DISABLE is recommended. If this parameter is set to ENABLE , the BSC6960 will allow sending of ICMP destination unreachable response messages for a long period of time. As a result, attackers may obtain the network topology of the BSC6960.

4.6.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

- Activating ACL-based packet filtering (manual ACL rule configuration in IPv4)
//Configuring an ACL
ADD ACL:ACLID=3000,ACLDISC="Acl Group is created";
//Configuring an ACL rule that uses IP as the protocol type and allows communication with the 192.168.5.0 network segment
ADD
ACLRULE:ACLID=3000,RULEID=10,ACTION=PERMIT,PT=IP,SIP="192.168.5.5",SWC="0.0.0.255",DIP="192.168.5.6",DWC="0.0.0.255",MDSCP=YES,DSCP=25;
//Configuring an ACL rule that uses ICMP as the protocol type and renders 192.168.5.5 pingable. Keeping this ACL rule poses security risks. It is good practice to delete this rule after the ping operation is complete.
ADD
ACLRULE:ACLID=3000,RULEID=11,ACTION=PERMIT,PT=ICMP,SIP="192.168.5.5",SWC="0.0.0.0",DIP="192.168.5.6",DWC="0.0.0.0",MDSCP=YES,DSCP=25;
//Configuring ACL-based packet filtering
ADD PACKETFILTERING: PACKETFILTERINGID=0, FM=ADV_ACL, ACLTYPE=IPV4, PT=INTERFACE, PORTID=60, ACLID=3000, MB=DENY;
//Activating alarm reporting for invalid packets exceeding a threshold
SET IPGUARD:INVALIDPKTCHKSW=ENABLE,INVALIDPKTALMTHD=2000;
- Activating other security functions
//Activating flood attack prevention
ADD FLOODDEFEND:FLDTYPE=ARP,DFDSW=ENABLE,DFDTHD=512,ALMSW=ENABLE;
//Activating ARP spoofing prevention
SET IPGUARD:ARPSPOOFCHKSW=ENABLE,ARPSPOOFALMTHD=120,ARPLRNSTRICTSW=ENABLE;
//Activating ICMP response attack prevention
SET TRANSFUNCTIONSW:ICMPPORTUNREACHABLESW=DISABLE;

Deactivation Command Examples

- Deactivating ACL-based packet filtering
//Removing the packet filtering policy
RMV PACKETFILTERING: PACKETFILTERINGID=0;
//Removing ACL rules
RMV ACLRULE: ACLID=3000, RULEID=10;
RMV ACLRULE: ACLID=3000, RULEID=11;
//Removing the ACL
RMV ACL: ACLID=3000;
//Disabling alarm reporting for invalid packets exceeding a threshold
SET IPGUARD: INVALIDPKTCHKSW=DISABLE;
- Deactivating other security functions
//Deactivating flood attack prevention. Deactivating flood attack prevention requires removing configurations for all types of attack packets. The example in this section is specific to ARP packets only.
RMV FLOODDEFEND:FLDTYPE=ARP;
//Deactivating alarm reporting for invalid packets exceeding a threshold
SET IPGUARD:INVALIDPKTCHKSW=DISABLE;
//Deactivating ARP spoofing prevention
SET IPGUARD:ARPSPOOFCHKSW=DISABLE, ARPLRNSTRICTSW=DISABLE;
//Deactivating ICMP response attack prevention
SET TRANSFUNCTIONSW: ICMPPORTUNREACHABLESW=ENABLE;

4.6.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

4.6.3 Activation Verification

The activation observation procedures in this section are applicable when network attacks exist.

ACL-based Packet Filtering

Step 1 Verify that permitted packets are properly received.

Use SCTP packets as an example. Run the **DSP SCTPLNK** command. If the value of **SCTP Link Status** is **Up** in the command output, the function is activated.

Step 2 Verify that denied packets are discarded.

Use a device whose packets should be denied to ping the interface IP address of the BSC6960. If the device receives a response indicating a ping failure, ACL-based packet filtering has been activated.

----End

Flood Attack Prevention

Run the **DSP FLOODDEFEND** command. If the value of **Number of Flood Dropped Packets** is not **0** in the command output, flood attack prevention is activated.

Invalid Packet Alarm Reporting

Run the **DSP INVALIDPKTINFO** command. If invalid packet information is available in the command output, invalid packet alarm reporting is successfully activated.

ARP Spoofing Prevention

Run the **DSP ARPSPOOFING** command. If IP address and MAC address information of ARP spoofed packets is available in the command output, ARP spoofing prevention is activated.

ICMP Response Attack Prevention

Check whether the peer device receives ICMP destination unreachable response packets. If no ICMP unreachable response packet is received, ICMP response attack prevention is activated.

4.6.4 Network Monitoring

None

5 Physical Port Security

5.1 Principles

5.1.1 Physical Port Security for the BSC6900/BSC6910 (GSM/UMTS)

To ensure port security of BSC6900/BSC6910, users need to disable unused ports (some ports are disabled by default, and some ports are enabled by default but their status is configurable) or enable authentication of ports. [Table 5-1](#) and [Table 5-2](#) describe physical port security measures taken by boards of the BSC6900/BSC6910.

Table 5-1 BSC6900 physical port security

Board	Physical Port	Port Type	Is Port Disabled by Default	Authentication Method	Port Function
OMUa/ OMUb	COM serial ports (COM0-ALM/ COM1-BMC)	DB9	No (port status is configurable)	OS authentication	Local commissioning
	One Ethernet port for commissioning	RJ45	No (port status is configurable)	Log in to OS: OS authentication Log in to and operate on the local maintenance terminal: OM authentication	Local commissioning
	Two service ports	RJ45	No	OM authentication	External service channel of the OM function

Board	Physical Port	Port Type	Is Port Disabled by Default	Authentication Method	Port Function
OMUc	One COM serial port	DB9	Yes (port status is configurable)	OS authentication	Local commissioning
	One Ethernet port for commissioning	RJ45	No (port status is configurable)	Log in to OS: OS authentication Log in to and operate on the local maintenance terminal: OM authentication	Local commissioning
	Two service ports	RJ45	No	OM authentication	External service channel of the OM function
SAUa	COM serial ports (COM0-ALM/COM1-BMC)	DB9	No	OS authentication	Local commissioning
	One Ethernet port for commissioning	RJ45	No	Log in to OS: OS authentication Log in to and operate on the local maintenance terminal: OM authentication	Local commissioning
	Two service ports	RJ45	No	OM authentication	External service channel of the OM function
SAUc	One COM serial port	DB9	Yes	OS authentication	Local commissioning
	One Ethernet port for commissioning	RJ45	No	Log in to OS: OS authentication Log in to and operate on the local maintenance terminal: OM authentication	Local commissioning
	Two service ports	RJ45	No	OM authentication	External service channel of the OM function

Board	Physical Port	Port Type	Is Port Disabled by Default	Authentication Method	Port Function
SCUa	One COM serial port	RJ45	Yes (the port cannot be enabled)	None	Commissioning by R&D and manufacturing personnel
	Ethernet ports (including 12 electrical ports) for inter-subrack connection	RJ45	MPS and remote main subracks: Yes EPS and TC extended subracks: Only ports 0 and 1 are enabled by default and cannot be disabled.	None	Inter-subrack connection
SCUb	One COM serial port	RJ45	Yes (the port cannot be enabled)	None	Commissioning by R&D and manufacturing personnel
	Ethernet ports (including 8 electrical ports and 4 optical ports) for inter-subrack connection	RJ45 (electrical port) SFP+ or LC/PC (optical port)	MPS and remote main subracks: Yes EPS and TC extended subracks: Only ports 0/1/8/9 are enabled by default.	None	Inter-subrack connection
XPUa/ XPUb/ XPUc	Four Ethernet ports on the panel	RJ45	The Ethernet ports on the panel of a serving BSC can be enabled by running MML commands to access CBS services.	The IPsec-capable external security gateway can be used for GSM authentication.	Connecting the base station controller to the CBS on a GSM network

Board	Physical Port	Port Type	Is Port Disabled by Default	Authentication Method	Port Function
SPUa/ SPUb/ SPUc	Four Ethernet ports on the panel	RJ45	The Ethernet ports on the RNC panel cannot be enabled by running MML commands. The ports are actually in the disabled state without MAC, VLAN, or IP address information.	None	Connecting the base station controller to the CBS on a UMTS network
GCUa/ GCGa/ GCUb/ GCGb	Two COM ports	RJ45	No	None	Inputting signals from external reference clock sources
Interface boards	Service port	Different services use different port types.	No	The IPsec-capable external security gateway can be used for authentication.	External service port of the base station controller

Table 5-2 BSC6910 physical port security

Board	Physical Port	Port Type	Is Port Disabled by Default	Authentication Method	Port Function
EOMUa/ EOMUb	One Ethernet port for commissioning	RJ45	No (port status is configurable)	Log in to OS: OS authentication Log in to and operate on the local maintenance terminal: OM authentication	Local commissioning
	Two service ports	RJ45	No	OM authentication	External service channel of the OM function
ESAUa/ ESAUb	COM serial ports (COM0-ALM/ COM1-BMC)	DB9	No	OS authentication	Local commissioning

Board	Physical Port	Port Type	Is Port Disabled by Default	Authentication Method	Port Function
	One Ethernet port for commissioning	RJ45	No	Log in to OS: OS authentication Log in to and operate on the local maintenance terminal: OM authentication	Local commissioning
	Two service ports	RJ45	No	OM authentication	External service channel of the OM function
SCUb	One COM serial port	RJ45	Yes (the port cannot be enabled)	None	Commissioning by R&D and manufacturing personnel
	Ethernet ports (including 8 electrical ports and 4 optical ports) for inter-subrack connection	RJ45 (electrical port) SFP+ or LC/PC (optical port)	MPS and remote main subracks: Yes EPS and TC extended subracks: Only ports 0/1/8/9 are enabled by default.	None	Inter-subrack connection
SCUc	Ethernet ports (including 12 optical ports) for inter-subrack connection	QSFP+, SFP, or LC/PC (optical port)	MPS and remote main subracks: Yes EPS and TC extended subracks: Only ports 0/1/2/3/8/9 are enabled by default.	None	Inter-subrack connection
GCUa/ GCGa/ GCUb/ GCGb/ GCUc/ GCGc	Two COM ports	RJ45	No	None	Inputting signals from external reference clock sources

Board	Physical Port	Port Type	Is Port Disabled by Default	Authentication Method	Port Function
Interface boards	Service port	Different services use different port types.	No	The IPsec-capable external security gateway can be used for authentication.	External service port of the base station controller

Table 5-3 describes the function of serial port command security provided in the BIOS phase.

Table 5-3 Serial port command security in the BIOS phase

Command	Description	Setting Notes	Application Scenario	Applicable Board
Ctrl+S	Used to start boot settings	Prohibited	This command is used when boot selection starts.	OMUa OMUb SAUa
Ctrl+C	Used to perform LSI configurations	Only used in special scenarios	This command is used only for setting RAID 1 on the OMU hard disks. For details about the setting, see <i>OMU Administration Guide</i> for the related product.	EOMUa EOMUb ESAUa ESAUb

5.1.2 Physical Port Security for the Base Station

The base station does not have an independent board for OM functions. To reduce unauthorized access and information disclosure, unused ports need to be disabled.

- Ethernet port for transmission
Transmission ports need to be disabled remotely.
- Port for commissioning
To perform local maintenance of a base station, operators connect the operation and maintenance terminal to ports for commissioning on the base station. Ports for commissioning are based on IP transmission, among which some TCP and UDP ports are enabled, possibly resulting in attacks on the base station.
To disable the ports for commissioning, users need to set the **LOCALETHPORT.SWITCH (LTE eNodeB, 5G gNodeB)** parameter to **DISABLE**.
If the base station is disconnected from the OSS, the disabled Ethernet port for commissioning is automatically enabled to ensure normal maintenance of

the base station. After the connection between the base station and the OSS is recovered, the port is automatically disabled. The actual port status can be queried by running the **DSP LOCALETHPORT** command.

After an Ethernet port for commissioning is enabled, the base station may be attacked. Therefore, the base station reports ALM-26294 High-Risk Function Enable to remind users to disable the local Ethernet port for commissioning in a timely manner.

- USB port

The USB port of the base station is used only as a commissioning Ethernet port and cannot be used for base station configuration and base station information export.

- Clock port or clock test port

The **BLK PHYPORT** command can be used to block a clock port, with the **PT** parameter set to **CLK**, or to block a clock test port, with the **PT** parameter set to **TST**.

A blocked clock port or clock test port can be unblocked by running the **UBL PHYPORT** command.

The status of a clock port or clock test port can be queried by running the **DSP PHYPORT** command.

- Outdoor BBU interconnection port

Outdoor BBUs use Ethernet ports for interconnection. An interconnection cable transmits sensitive data (including control-plane data and user-plane data) in plaintext, which is vulnerable to unauthorized decryption. An interconnection port can be blocked by running the **BLK PHYPORT** command with **PT** set to **HEI**.

When the base station requires interconnection and the peer equipment has been verified as authorized, the interconnection port can be unblocked by running the **UBL PHYPORT** command.

The port status can be queried by running the **DSP PHYPORT** command.

- CPRI_E/CPRI_O port on the RHUB

The RHUB connects to a pRRU through a CPRI_E/CPRI_O port, and data is encapsulated using CPRI and transmitted using network cables or optical fibers. A CPRI_E/CPRI_O port can be disabled by running the **BLK CPRIPORT** command.

After a CPRI_E/CPRI_O port is disabled, users need to run the **UBL CPRIPORT** command to enable it if the RHUB needs to be connected to a valid pRRU through this port.

Users can run the **DSP CPRIPORT** command to query the status of this port.

5.1.3 Physical Port Security for the BSC6960

The BSC6960 does not have an independent board for OM functions. The OM functions are performed on the CPMU. To reduce unauthorized access and information disclosure, unused ports need to be disabled.

- Ethernet port for transmission

Transmission ports need to be disabled remotely.

- Port for commissioning

To perform local maintenance of a BSC6960, operators connect the operation and maintenance terminal to ports for commissioning on the BSC6960. Ports for commissioning are based on IP transmission, among which some TCP and UDP ports are enabled, possibly resulting in attacks on the base station.

To disable the ports for commissioning, users need to set the **LOCALETHPORT.SWITCH** parameter to **DISABLE**.

If the BSC6960 is disconnected from the OSS, the disabled Ethernet port for commissioning is automatically enabled to ensure normal maintenance of the BSC6960. After the connection between the BSC6960 and the OSS is recovered, the port is automatically disabled. The actual port status can be queried by running the **DSP LOCALETHPORT** command.

- USB port

The USB port of the BSC6960 is used only for commissioning purposes and cannot be used for configuration and BSC6960 information export.

5.1.4 USB Port Security

On GSM and UMTS networks, installation, switchovers, or recovery can be implemented on the BSC6900/BSC6910 by connecting it to a USB storage device through a USB port. Disclosure of configuration files in the USB storage device poses significant security threats. The configuration files in the USB storage device must be encrypted.

When a software package for the BSC6900/BSC6910 is released on GSM and UMTS networks, the software package is digitally signed. Therefore, during software installation through a USB port, software integrity check is performed to ensure that the installed software is issued by Huawei and not tampered with.

When the BSC6900/BSC6910 is properly running on GSM and UMTS networks, no features or functions dependent on USB ports are applied. You are advised to run the **SET OMUPORT** command to disable USB ports to prevent potential risks.

5.2 Network Analysis

5.2.1 Benefits

This function reduces the risks of unauthorized device access and information leakage.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations (macro base stations)
UMTS	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910. Only NR TDD is supported.

Boards

No requirements

RF Modules

This function does not depend on RF modules.

5.3.4 Others

None

5.4 Operation and Maintenance

5.4.1 When to Use

Engineering guidelines for physical port security apply only to base stations and include the following:

- Transmission port
It is recommended that the transmission port be always enabled.
- Port for commissioning
It is recommended that the port for commissioning be disabled after the NE is connected to the MAE. It is recommended that the alarm detection switch be turned on.
- Clock port or clock test port
It is recommended that a clock port or clock test port be disabled if not needed.
- Port for interconnection
It is recommended that the port for interconnection be disabled if outdoor BBUs are not interconnected.
- CPRI_E/CPRI_O port
It is recommended that a CPRI_E/CPRI_O port on an RHUB be disabled if it is not used for connection with pRRUs.

5.4.2 Data Configuration

5.4.2.1 Data Preparation

Before configuring physical port security, prepare information about ports that do not carry any services and those that are not used temporarily. For details about ports that must be enabled for a base station, see 3900 & 5900 series base station communication matrix in *3900 & 5900 Series Base Station Product Documentation*.

Parameters must be set in the **PORTSECURITY** MO to disable a transmission port. The cabinet number, subrack number, slot number, and subboard type in this MO are derived from the network plan (negotiation not required) and must be set to those of the board to be enabled with packet filtering. [Table 5-4](#) describes the key parameters.

Table 5-4 Data to be prepared before disabling a transmission port

Parameter Name	Parameter ID	Setting Notes
Port Type	PORTSECURITY.PT (<i>5G gNodeB, LTE eNodeB</i>)	Set this parameter based on the network plan. This parameter specifies the type of port to which port security applies.
Port No.	PORTSECURITY.PN (<i>5G gNodeB, LTE eNodeB</i>)	This parameter specifies the number of a port.
Enable Flag	PORTSECURITY.SWITCH (<i>5G gNodeB, LTE eNodeB</i>)	This parameter specifies whether to disable a port for security purposes. Port security cannot be enabled at a port that carries an upper-layer object. It is recommended that the transmission ports that do not carry services be disabled to reduce the probability of system attacks.

Parameters must be set in the **LOCALETHPORT** MO to disable an Ethernet port for commissioning. [Table 5-5](#) describes the key parameters.

Table 5-5 Data to be prepared before disabling an Ethernet port for commissioning

Parameter Name	Parameter ID	Setting Notes
Enable Flag	LOCALETHPORT.SWITCH (<i>5G gNodeB, LTE eNodeB</i>)	This parameter specifies whether to enable an Ethernet port for commissioning. If this parameter is set to DISABLE, the Ethernet port for commissioning is disabled and local commissioning cannot be performed. It is recommended that the commissioning Ethernet port be disabled to reduce the probability that attackers use this port to locally connect to the base station.

Parameter Name	Parameter ID	Setting Notes
Alarm Switch	LOCALETHPORT.ALMSW <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter specifies whether an alarm is reported when the local Ethernet port for commissioning is enabled. If this parameter is set to ENABLE , the alarm will be reported. If this parameter is set to DISABLE , the alarm will not be reported. It is recommended that the alarm switch for Ethernet ports for commissioning be enabled. After an Ethernet port for commissioning is enabled, an alarm is reported in time to remind users to disable the port.

The following table describes key parameters that must be set in the **PHYPORT** MO to disable a clock port or clock test port. The cabinet number, subrack number, slot number, and subboard type in this MO must be set to those of the board providing the clock port or clock test port. It is recommended that a clock port or clock test port be disabled.

Table 5-6 Data to be prepared before disabling a clock port or clock test port

Parameter Name	Parameter ID	Setting Notes
Port Type	PHYPORT.PT <i>(5G gNodeB, LTE eNodeB)</i>	Set this parameter based on the network plan. This parameter specifies the type of port to which port security applies. The value CLK indicates a clock port and the value TST indicates a clock test port.
Port No.	PHYPORT.PN <i>(5G gNodeB, LTE eNodeB)</i>	This parameter specifies the number of a port.

The following table describes key parameters that must be set in the **PHYPORT** MO to disable an interconnection port. The cabinet number, subrack number, slot number, and subboard type in this MO are derived from the network plan (negotiation not required) and must be set to those of the board to be enabled with packet filtering.

In outdoor BBU interconnection scenarios, it is recommended that the interconnection port that does not carry services be disabled to reduce the probability of system attacks.

Table 5-7 Data to be prepared before disabling a port for interconnection

Parameter Name	Parameter ID	Setting Notes
Port Type	PHYPORT.PT (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan. This parameter specifies the type of port to which port security applies. The value HEI indicates an interconnection port of an outdoor BBU.
Port No.	PHYPORT.PN (5G gNodeB, LTE eNodeB)	This parameter specifies the number of a port.

The following table describes key parameters that must be set in the **CPRIPORT** MO to disable a CPRI_E/CPRI_O port on the RHUB. The cabinet number, subrack number, and slot number in this MO are used to specify the RHUB that provides the port and are derived from the network plan (negotiation not required).

It is recommended that the CPRI_E/CPRI_O ports be enabled on the RHUB. If a CPRI_E/CPRI_O port on the RHUB is disabled, services on the board carried on the RRU chain/ring connected to this port will be interrupted and an alarm will be reported.

Table 5-8 Data to be prepared before disabling a CPRI_E/CPRI_O port on the RHUB

Parameter Name	Parameter ID	Setting Notes
Port No.	CPRIPORT.OPTN (5G gNodeB, LTE eNodeB)	This parameter specifies the number of a port.

NOTE

If the RHUB and pRRUs work in branch load sharing topology, both of the two CPRI_E/CPRI_O ports connected to each pRRU must be disabled. For details about the branch load sharing topology, see *CPRI MUX (LampSite)* in *3900 & 5900 Series Base Station Product Documentation*.

5.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Disabling a transmission port
SET PORTSECURITY: SN=6, SBT=BASE_BOARD, PT=ETH, PN=0, SWITCH=DISABLE;
//Disabling a port for commissioning
SET LOCALETHPORT: SWITCH=DISABLE,ALMSW=ENABLE;
//Disabling a clock port
BLK PHYPORT: CN=0, SRN=0, SN=7, PT=CLK, PN=0;
//Disabling a clock test port
BLK PHYPORT: CN=0, SRN=0, SN=7, PT=TST, PN=0;
//Disabling a port for interconnection
BLK PHYPORT: CN=0, SRN=0, SN=6, PT=HEI, PN=0;
//Disabling a CPRI_E/CPRI_O port on the RHUB
BLK CPRIPORT: CN=0, SRN=60, SN=0, OPTN=2;
```

Deactivation Command Examples

```
//Enabling a transmission port
SET PORTSECURITY: SN=6, SBT=BASE_BOARD, PT=ETH, PN=0, SWITCH=ENABLE;
//Enabling a port for commissioning
SET LOCALETHPORT: SWITCH= ENABLE;
//Enabling a clock port
UBL PHYPORT: CN=0, SRN=0, SN=7, PT=CLK, PN=0;
//Enabling a clock test port
UBL PHYPORT: CN=0, SRN=0, SN=7, PT=TST, PN=0;
//Enabling a port for interconnection
UBL PHYPORT: CN=0, SRN=0, SN=6, PT=HEI, PN=0;
//Enabling a CPRI_E/CPRI_O port on the RHUB
UBL CPRIPORT: CN=0, SRN=60, SN=0, OPTN=2;
```

5.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3 Activation Verification

Transmission Port Security

- Step 1** Run the **DSP PORTSECURITY** command to query the port status. Verify that the value of **Enable Flag** is **Disable** in the command output.
- Step 2** Run the **ADD DEVIP** command to add an IP address to the disabled port. If the system displays a message indicating that port is disabled, the security function has been activated successfully on this port.

----End

Commissioning Port Security

- Step 1** Connect a PC to the port for commissioning on the base station.

- Step 2** On the PC, ping the IP address of the port. If the ping operation fails, the security function has been activated successfully on the commissioning port.

----End

Clock Port or Clock Test Port Security

Run the **DSP PHYPORT** command to query the port status. If **Administrative State** is set to **Blocked** for the port with **Port Type** set to **CLK** or **TST** in the command output, clock port or clock test port security has been activated.

Interconnection Port Security

Run the **DSP PHYPORT** command to query the port status. If **Port Type** is **HEI** and **Administrative State** is **Blocked** in the command output, the security function has been activated successfully on the interconnection port.

CPRI_E/CPRI_O Port Security

Run the **DSP CPRIPORT** command to query the port status. If **Administrative Status** is **Blocked** in the command output, the security function has been activated successfully on the CPRI_E/CPRI_O port.

5.4.4 Network Monitoring

None

6 Other Functions

6.1 Physical Security

Physical security refers mainly to physical security measures.

6.1.1 Physical Security for the Base Station

- The following measures are recommended for protection of physical security of indoor macro base stations:
 - Devices are placed in equipment rooms with door locks and other security facilities.
 - Environment monitoring units and sensors are configured to monitor the equipment room environment. When a threshold is reached, related alarms, such as water damage alarms and smoke alarms, are reported.
- Outdoor macro base stations are secured in cabinets with door locks and door status alarms.

If sensors such as the door status sensor, water sensor, and smoke sensor are not configured, the site environment security status may not be monitored in a timely manner, which increases physical security risks.

6.1.2 Physical Security for the Base Station Controller (GSM/UMTS)

The following measures are recommended for protection of physical security of base station controllers:

- Devices are placed in equipment rooms with door locks and other security facilities.
- Environment monitoring units and sensors are configured to monitor the equipment room environment. When a threshold is reached, related alarms, such as water damage alarms and smoke alarms, are reported.

If sensors such as the door status sensor, water sensor, and smoke sensor are not configured, the site environment security status may not be monitored in a timely manner, which increases physical security risks.

6.2 OS Security

6.2.1 OS Security of the Base Station/BSC6960

Base station software encapsulates the OS in components. [Table 6-1](#) lists the OSs and security measures supported by the base station.

Table 6-1 OSs and security measures supported by the base station

Product Model	OS	Description	OS Hardening	OS Patch
eGBTS	Real-time operating system (RTOS) of the base station	The embedded OS is encapsulated in the base station software.	Supported	Supported
NodeB			Supported	Supported
eNodeB		Hardening has been performed for the OS before delivery.	Supported	Supported
gNodeB			Supported	Supported
Multimode base station		The OS patch is upgraded with the base station software.	Supported	Supported
Note: For details about base station RTOS security, see <i>RTOS Security</i> of SingleRAN.				

BSC6960 software encapsulates the OS in components. [Table 6-2](#) lists the OSs and security measures supported by the BSC6960.

Table 6-2 OSs and security measures supported by the BSC6960

Product Model	OS	Description	OS Hardening	OS Patches
BSC6960	RTOS	The embedded OS is encapsulated in the software. Hardening has been performed for the OS before delivery. The OS patch is upgraded with the software.	Supported	Supported
Note: For details about RTOS security, see <i>RTOS Security</i> of SingleRAN.				

6.2.1.1 OS Hardening

If the base station/BSC6960 OS has security vulnerabilities and potential risks, these vulnerabilities may be exploited by local or remote attackers to impose security threats on the OS and related software, thereby affecting normal system operation.

In view of the foregoing security risks, Huawei base station/BSC6960 OSs are hardened before delivery. The solutions cover network access, network security, and system services to improve antivirus and anti-attack capabilities, system reliability, and the service quality of the entire network.

The OS hardening solutions include the following functions:

- Disabling unnecessary services
- Restricting access to files and directories
- Authorizing system access
- Managing users
- Recording operation logs
- Detecting system malfunctions
- Protecting OS integrity

6.2.1.2 OS Patches

- The full OS patch is released periodically (once a year). The latest base station/BSC6960 software will include the OS patch.
- After the base station/BSC6960 is delivered, the OS patch is upgraded with the base station/BSC6960 software. The OS cannot be upgraded independently. The base station/BSC6960 OS is invisible to users.

6.2.2 OS Security of the BSC6900/BSC6910 (GSM/UMTS)

This section focuses on OS security of the base station controller.

The OMU acts as the bridge for information exchange between the base station controller and other NEs. [Table 6-3](#) describes the OSs and security measures (system hardening and patches) supported by the OMUs of base station controllers.

Table 6-3 OSs and security measures supported by base station controllers

Product Model	OS	OS Hardening	OS Patch
BSC6900	Dopra Linux	Supported	Supported
BSC6910	Dopra Linux and Euler OS		
Note: For details about Dopra Linux security, see <i>Dopra Linux OS Security</i> in GBSS feature documentation or RAN feature documentation.			

6.2.2.1 OS Hardening

The OS and related software have security holes and potential risks, which may be exploited by local or remote attackers, thereby affecting the normal operation of the OS.

Huawei provides OS hardening solutions. These solutions cover network access, network security, system service, and system installation to improve antivirus and anti-attack capabilities, system reliability, and the service quality of the entire network.

The OS hardening solutions include the following functions:

- Disabling unnecessary services
- Reinforcing Secure Shell (SSH) services
- Restricting access to files and directories
- Authorizing system access
- Managing user passwords
- Recording operation logs
- Detecting system malfunctions

Table 6-4 describes the security hardening solutions of different OSs:

Table 6-4 OS hardening solutions

OS	Security Hardening Solution
Dopra Linux	Dopra Linux and Euler Linux are Huawei-developed OSs. They have been reinforced before delivery and therefore do not require additional hardening.
Euler OS	

6.2.2.2 OS Patches

The latest patch packages have been installed on the base station controller before delivery. Dopra Linux and Euler OS patches are released at least once a year.

NOTE

For details about OS patches for a specific product version, see the corresponding release notes.

Users can obtain the latest patch packages by going to <https://support.huawei.com> and choosing **Tools > Mini-tool Software > Wireless Product Line > Universal OS Patches**. Users can also contact Huawei technical support engineers to obtain the patch packages.

Users can install patches for a Huawei base station controller in either of the following modes:

- Local patch installation

An OM engineer must log in to the OMU OS to install OS patches for only one base station controller at a time.

- Remote patch installation

An OM engineer uses Huawei network management software to install OS patches for multiple base station controllers.

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*.

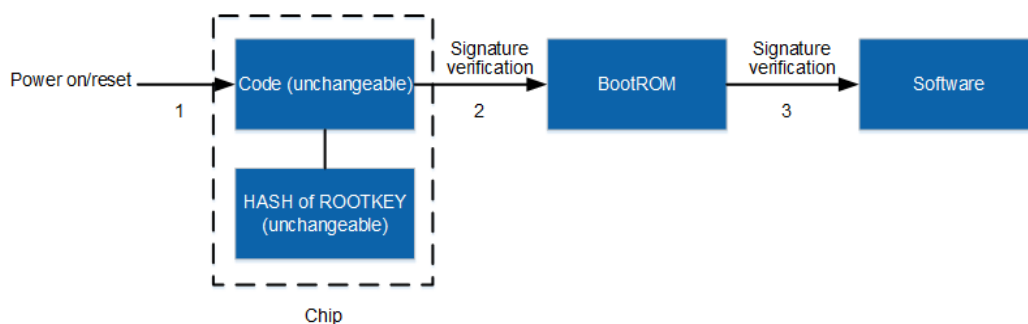
- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

6.3 Base Station Security Environment

6.3.1 Secure Boot

After a base station is powered on or reset, the fixed code in the chip loads the next-level BootROM and software in sequence. The signature must be verified before the BootROM and software are executed, preventing malicious software from running, as shown in [Figure 6-1](#). The key revocation file version of the BootROM can be queried by running the **DSP CHIPVER** command.

Figure 6-1 Secure boot process



The software signature uses the PKI-CMS signature mechanism. For details, see *OM Security*.

The following boards support this function:

- UMPTg, UMPTga, UMPTha, and UMPThb series boards
- UBBPg and UBBPh series boards
- AAUs that support eCPRI in SRAN16.1 and later versions

6.3.2 Secure Storage

In secure storage, encryption keys for sensitive data (such as private keys and passwords) on base stations are layered. Confidentiality of upper-layer keys depends on lower-layer keys. Root keys are at the bottom layer. Root key materials are protected using mechanisms such as secure software storage and access control.

The base station master key is protected by the root key and provides confidentiality protection for the working keys. The master key supports automatic update, and the update period is specified by running the **SET KEYPOLICY** command. The system checks every early morning whether the key update period has arrived and updates the key if the period has arrived. The automatic update period of the master key can be queried by running the **LST KEYPOLICY** or **DSP KEYPOLICY** command. The latest update time of the master key can be queried by running the **DSP KEYPOLICY** command.

Administrators can run the **UPD KEYMATERIAL** command to update the key material for encrypting local sensitive data. It is recommended that the administrator periodically update the system key material based on the management policy. If the base station key material is not updated for a long time, attackers may easily obtain the key for encrypting local sensitive data and obtain the plaintext of the sensitive data.

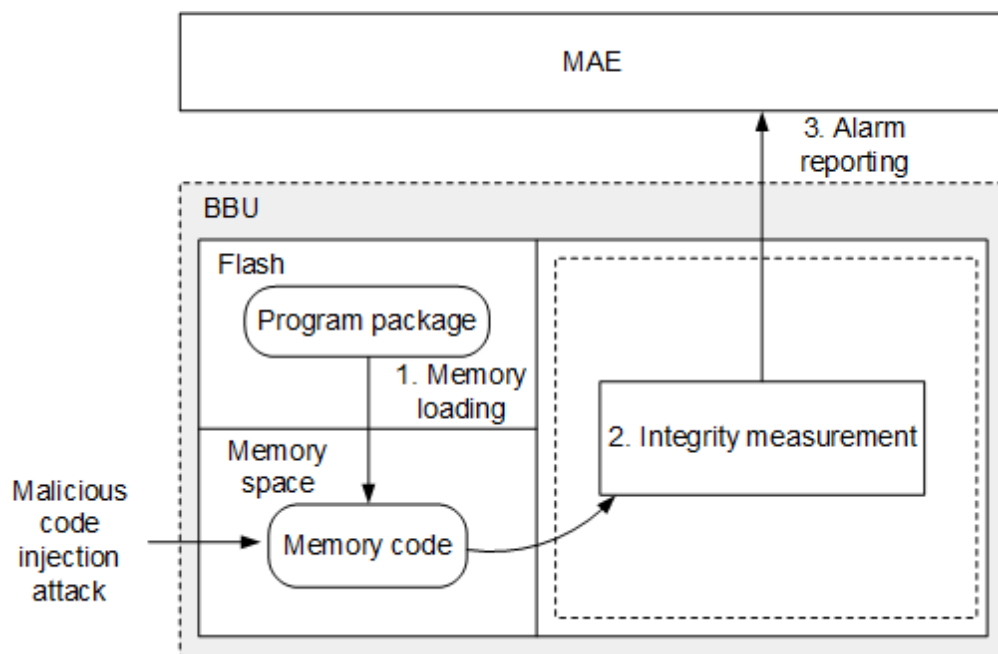
The following boards support storing root key materials inside the chip so that the materials are inaccessible from outside, ensuring the security of sensitive data stored in the system.

- UMPTe, UMPTg, UMPTga, UMPTha, and UMPThb series boards
- UBBPg and UBBPh series boards

6.3.3 Memory Code Integrity Measurement

During long-term running, program code loaded to the memory may be maliciously tampered with, for example, malicious code may be injected by exploiting external communication protocol vulnerabilities. To cope with this situation, base stations provide a memory code integrity measurement function, which periodically checks the runtime integrity of the memory code. If the code is tampered with, ALM-26203 Board Software Program Error is immediately reported to the OSS, allowing the abnormal event to be handled in time, as shown in [Figure 6-2](#).

Figure 6-2 Memory code integrity measurement process



The following boards support this function:

- UMPTe, UMPTg, UMPTga, UMPTha, and UMPThb series boards
- UMDUb
- UBBPg and UBBPh series boards

6.4 Base Station Self-Check Upon Startup

During the startup, a base station automatically checks its hardware and software and saves the self-check results. The base station also provides MML commands for querying the self-check results.

- Hardware self-check involves hardware components, such as the memory and flash. You can run the **DSP HWOLTSTRESULT** command to query self-check results.
- Software self-check provides self-checks for kernel and processes. You can run the **DSP SWTSTRESULT** command to query self-check results.

6.5 Base Station Process Auditing

Malware may be planted into a running base station. If the base station detects malware, the base station marks the process status of the malware as unknown.

You can run the **DSP PROCESSINFO** command to obtain the process status.

6.6 Key File Integrity Monitoring

Technical Description

When an NE is operating, key files may be maliciously modified. As a result, the system is operating unsafely or resources are illegally utilized. The key file integrity monitoring function automatically checks whether a key file has been tampered with and reports the check result to the MAE.

The system calculates the hash value of key files as the baseline at the first startup after an upgrade. When a key file is checked, its hash value is calculated and compared with the baseline. If the two hash values are inconsistent, the system determines that the key file has been modified. Integrity check reports compare the check items at the check time and at the time when the baseline is obtained. If a key file is modified again to the baseline before it is checked, the file is regarded as not modified.

If a modified key file is detected, whether the modification is normal (such as system upgrade or system maintenance) must be determined. If the modification is normal, choose **Security > Integrity Monitoring** on the MAE-Access, open the **Query and Set NE Check Information** tab page, and update the baseline.

Key file check involves the following check items:

- Service configuration files: files stored on the operating system of an NE for service configurations
- Audit logs: logs for audit

NOTE

- For a base station or USU, it is recommended that maintenance personnel update the baselines of service configuration check items in time after normally modifying key materials.
- A UMPT/UMDU/MDUC/GTMUc must be configured on the base station side. This function has no hardware requirement on the base station controller side.

Check Methods

The following two check methods are provided:

- Periodic check
Key files are checked once a day based on the specified check time and check items.
This can be achieved by choosing **Security > Integrity Monitoring** and open the **Query and Set NE Check Information** tab page on the MAE-Access.
- Immediate check
Integrity check can be performed on specified check items of an NE at any time.
This can be achieved by choosing **Security > Integrity Monitoring** and open the **Check Result** tab page on the MAE-Access.

It is recommended that periodic check be set on the OSS to detect malicious modification of key NE files in a timely manner.

Check Reports

Integrity check reports of key files consist of an overview (including the NE name, check items, and number of modified files) and details about modified files (including the check item and file name).

This can be achieved by choosing **Security > Integrity Monitoring** and open the **Check Result** tab page on the MAE-Access.

7 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

- *GBTS Equipment and OM Security* in GBSS feature documentation
- *Dopra Linux OS Security* in GBSS feature documentation or RAN feature documentation
- *Euler OS Security* in GBSS feature documentation or RAN feature documentation
- *IPv4 Transmission*
- *IPv6 Transmission*
- *OM Security*
- *RTOS Security*
- Documents in *3900 & 5900 Series Base Station Product Documentation*:
 - *CPRI MUX (LampSite)*
 - *3900 & 5900 Series Base Station Commissioning Guide*
 - *3900 & 5900 Series Base Station Communication Matrix*
- *OMU Administration Guide*
- *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in *eRAN Feature Documentation*